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Technical Bulletin



Technical Bulletin Unique Identifier	DOC2855227
Title	One new method of RF Maximum power calibration, RF amp and UPM functional check for the MR product with CSA
Subject	Service Note – One new method of RF Maximum power calibration, RF amp and UPM functional check for the MR product with CSA
Applicability (e.g., affected product, process)	All MR 1.5T products with CSA
Date	29th-May-2023
Relevant Product Business (select one)	☒ Imaging ☐ Patient Care Solutions ☐ Ultrasound ☐ Other: _____
Originating Modality	MR
Details	Apply 5307511-4/-5 kit in RF Maximum power calibration, RF amp and UPM functional check for the product with CSA.
Tracking Number	N/A
Affected Service Publications	Optima MR360/Brivo MR355/Optima MR360 Advance/Brivo MR355 Inspire 1.5T System Service Method Class A: 6336161-3EN Class C: 6336163-3EN Rev.4, Class M: 6336165-3EN Rev.4 Signa Creator/Explorer 1.5T System Service Manual Class A: 556356-2EN Rev.10, Class C: 5756355-2EN Rev.11, Class M: 5756347-2EN Rev.11 Signa MR 355/MR 360 1.5T System Service Method Class A: 5856356-3EN Rev.6, Class C: 5856355-3EN, Rev.7 Class M: 5856347-3EN Rev.6 Signa MR 380 1.49T System Service Manual Class A: 5856357-8EN Rev.3, Class C: 5856358-2EN Rev.2, Class M: 5856359-8EN Rev.2
Resolution	See attached
Technical Bulletin Owner	Qin Yu (SSO: 503343456)

(This Service Note is classified as Class A)

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Effectivity:

All MR 1.5T products with CSA

Overview:

Since the RF Power Measurement Kit (5307511-3) used by 1.5T SIGNA Artist and 1.5T SIGNA Voyager for **RF Body and Head Maximum Power Setup, Calibration and RAP/UPM Functional check** was upgraded to -4 and -5, the setting for measuring 10kw Body Output was added, RF Power Measurement Kit (5307511-4/-5) can be applied to CSA 10kw Body Channel as one new method of **RF Body and Head Maximum Power Setup, Calibration and RAP/UPM Functional check**.

This Service Note introduces the procedure of applying RF Power Measurement Kit (5307511-4/-5) as one new method for **RF Maximum power calibration, RF amp and UPM functional check** in 1.5T Signa MR 355/360, 1.49T Signa MR 380, 1.5T Optima MR360/Brivo MR355, 1.5T Optima MR360 Advance/Brivo MR355 Inspire, 1.5T Signa Creator/Explorer.

RF Power Measurement Kit (5307511-3) does not include the testing points of 10KW; thus, it shall not be used on 10kw Transmit chain systems.

Prerequisites:

Table 1. Personnel requirements

Required persons	Preliminary requirements	Procedure	Finalization
1	-	60 minutes	5 minutes

Table 2. Tools and test equipment

Item	Quantity	Effectivity	Part number	Manufacturer
RF power measurement kit (use one of the following kits):	1	-	EITHER 5307511-4, -5 (Bird wattmeter)	-
12-inch Crescent Wrench	1	-	-	-
Small Flat Head Screwdriver	1	-	-	-
Digital Multimeter	1	-	46-194427 P284	-

Table 3. Safety

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DANGER

FATAL ELECTRIC SHOCK HAZARD!
THE RF AMPLIFIER IS NOT DISABLED.
TO PREVENT FATAL ELECTRIC SHOCK, ENSURE THE RF AMPLIFIER IS DISABLED WHILE CONFIGURING THE CALIBRATION TOOLS.

WARNING

ELECTRICAL/RF BURN HAZARD!
RF BURNS Can Occur AS A RESULT OF DISCONNECTING HELIAX CABLES WHILE THE SYSTEM IS SCANNING.
PREVENT POSSIBLE RF BURNS WHEN DISCONNECTING HELIAX CABLES FROM THE RF AMPLIFIER BY VERIFYING THAT THE SYSTEM IS NOT MANUALLY PRESCANNING OR SCANNING. VERIFY THAT THE SCAN DESKTOP ICON DISPLAYS THE "IDLE" MESSAGE AND THE RF ENABLE SWITCH IS IN THE "OFF" POSITION.

CAUTION

POSSIBLE EQUIPMENT DAMAGE!
Operating the RF amplifier with an improper load can cause permanent damage to the amplifier.
Do not operate the RF amplifier with an improper load connected to its output.

CAUTION

POSSIBLE EQUIPMENT DAMAGE!
Hardware and phantoms left in the magnet bore can cause equipment damage.
Prevent coil and associated switch damage by removing all phantoms and hardware (such as head coil or surface coil) from the magnet bore.

CAUTION

POSSIBLE EQUIPMENT DAMAGE!
If the wattmeter sensor is connected or disconnected while the meter is ON, equipment damage can occur.
Do not connect/disconnect the wattmeter sensor while the meter is ON. Always turn the meter OFF before plugging/unplugging sensor.

Table 4. Required conditions

Condition	Reference	Effectivity
Before beginning this procedure, confirm that the exciter has been on for at least 20 minutes.	-	-

Table 5. Calibration Steps: CSA

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Step	Description
1	Disabling T/R, DD, and RF Input for Body Mode
2	Setting up RF Coupler for Body Calibration
3	Wattmeter and Dummy Load Setup for Body Channel (CSA)
4	Configuring the wattmeter for body channel
5	Setting up Bird 5000-EX Wattmeter to Measure Forward Body Maximum Power
6	Setting up Bird 5000-XT Wattmeter to Measure Forward Body Maximum Power
7	Setting up Bird TT800V Wattmeter to Measure Forward Body Maximum Power
8	Adjusting RF amplifier output power for body channel
9	RAP/UPM Functional Check for Body Mode
10	Disabling T/R, DD, and RF Input for Head Mode
11	Setting up RF Coupler for Head Calibration
12	Wattmeter and Dummy Load Setup for Head Channel (CSA)
13	Configuring the wattmeter for Head channel
14	Setting up Bird 5000-EX Wattmeter to Measure Forward Head Maximum Power
15	Setting up Bird 5000-XT Wattmeter to Measure Forward Head Maximum Power
16	Setting up Bird TT800V Wattmeter to Measure Forward Head Maximum Power
17	Adjusting RF amplifier output power for Head channel
18	RAP/UPM Functional Check for Head Mode
19	Finalization

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Procedure:

1. Disabling T/R, DD, and RF Input for Body Mode

DANGER

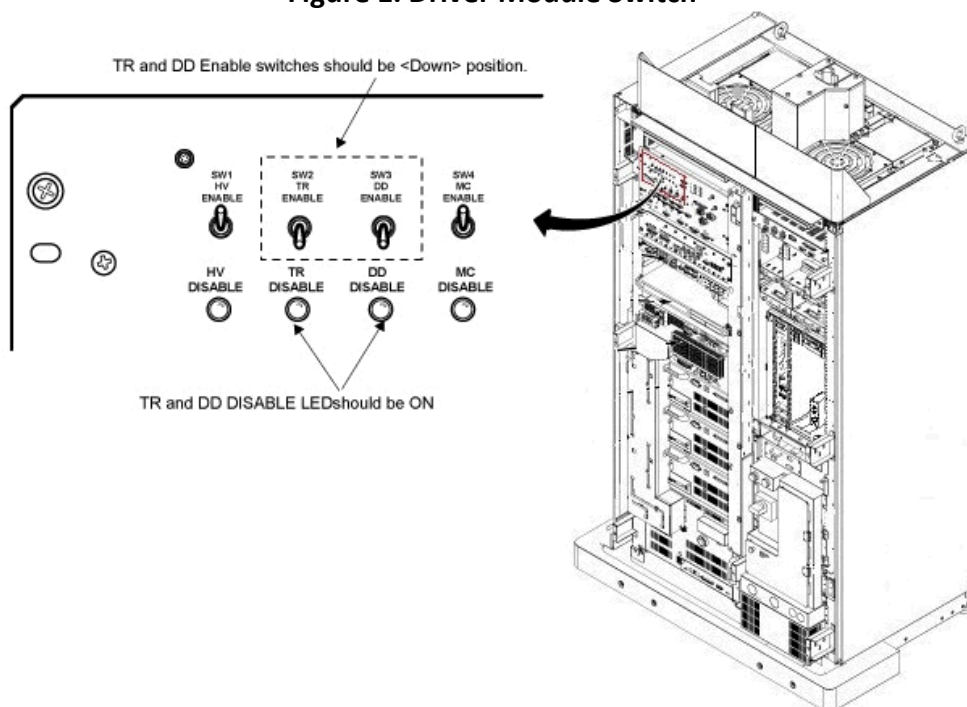
FATAL ELECTRIC SHOCK HAZARD.

the RF Amplifier is not disabled.

to prevent fatal electric shock, ensure the RF Amplifier is disabled while configuring the calibration tools.

- 1) Verify that the system is idle, and all coils have been removed from the magnet bore.
- 2) Remove the front cover of System Cabinet.
- 3) On the front of the Driver Module Lite, disable T/R error reporting by setting the TR and DD switches to the DISABLE (disable faults) position.

Figure 1. Driver Module Switch



NOTE: HV should remain set to Enable.

- 4) Place the RF Enable switch on the DTX Exciter, located in the top of the System cabinet, into the **Disable (down)** position.

Figure 2. RF ENB Switch



2. Setting up RF Coupler for Body Calibration

- 1) Insert the **50kW** element into the **Forward** sensor port (green 4300A374-2). Lock the element in place using the locking tab on the sensor. The Reflected port is not used during this procedure.

Placing the 5000W element (green 4300A374-5) into the **Reflected** port is just a safety measure and is not used during body calibration.

NOTE: Make sure the arrows on the elements point in opposite directions.

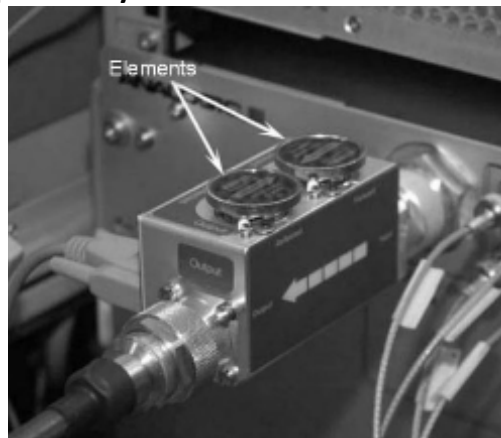
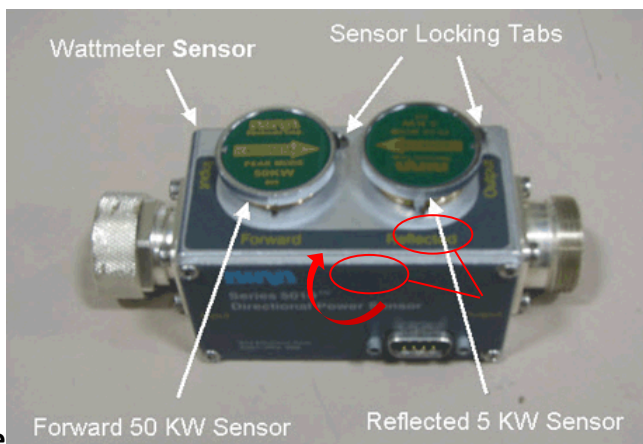
NOTE: Make sure that the two elements are plugged into the port and locked with buckles, otherwise it may cause inaccurate readings or no value display on the power meter.

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Figure 3. RF Coupler Setup for Body Mode



- 2) Use the DVM to measure the resistance of input to the dummy load. Confirm that the resistance measures 50 ohms +/- 5 ohms.

Figure 4. Measure dummy load input resistance



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NOTE: If the dummy load is out of spec, return the RF kit for repair. Do NOT proceed with the RF amplifier output calibration.

3. Wattmeter and Dummy Load Setup for Body Channel (CSA)

DANGER

FATAL ELECTRIC SHOCK HAZARD!

THE RF AMPLIFIER IS NOT DISABLED.

TO PREVENT FATAL ELECTRIC SHOCK, ENSURE THE RF AMPLIFIER IS DISABLED WHILE CONFIGURING THE CALIBRATION TOOLS.

WARNING

ELECTRICAL/RF BURN HAZARD!

RF BURNS Can Occur AS A RESULT OF DISCONNECTING HELIAX CABLES WHILE THE SYSTEM IS SCANNING.

PREVENT POSSIBLE RF BURNS WHEN DISCONNECTING HELIAX CABLES FROM THE RF AMPLIFIER BY VERIFYING THAT THE SYSTEM IS NOT MANUALLY PRESCANNING OR SCANNING. VERIFY THAT THE SCAN DESKTOP ICON DISPLAYS THE "IDLE" MESSAGE AND THE RF ENABLE SWITCH IS IN THE "OFF" POSITION.

NOTICE

Possible equipment damage. Do not connect/disconnect the wattmeter sensor while the meter is ON. Always turn the meter OFF before plugging/unplugging the sensor.

- 1) Verify that the scanner is idle. Confirm that the RF ENB switch on the DTX1 exciter is in the Disable (down) position.
- 2) Disconnect the body output cable from **J4** on the front panel of the RF amplifier.
- 3) Connect the input of the coupler to **J4** on the front panel of the RF amplifier.

Figure 5. Connection of RF Coupler to Body Output

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**WARNING****ELECTRIC SHOCK RISK!**

Failure to connect the dummy load correctly could result in electric shock. pay close attention to the connections to the dummy load.

- 4) Connect the output of the RF coupler to the dummy load.

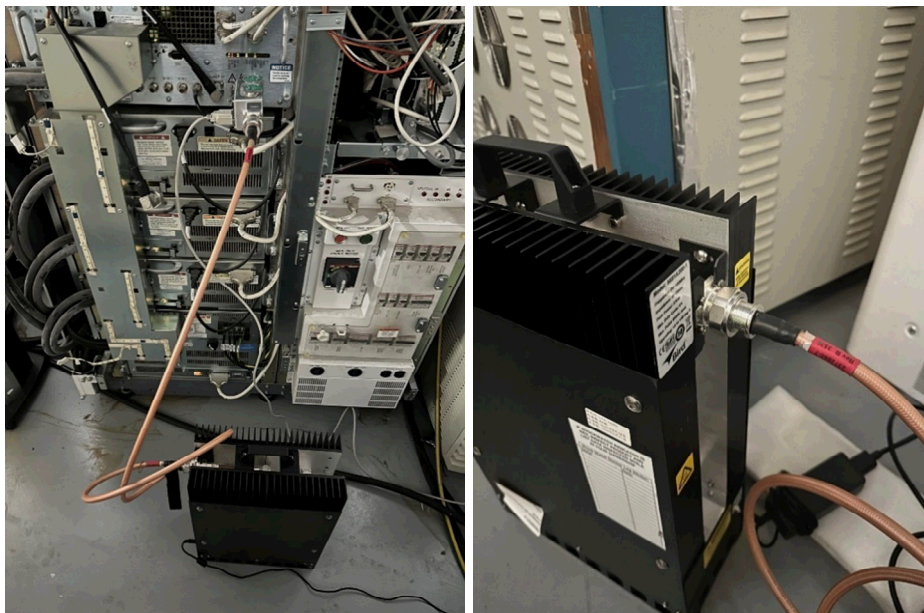
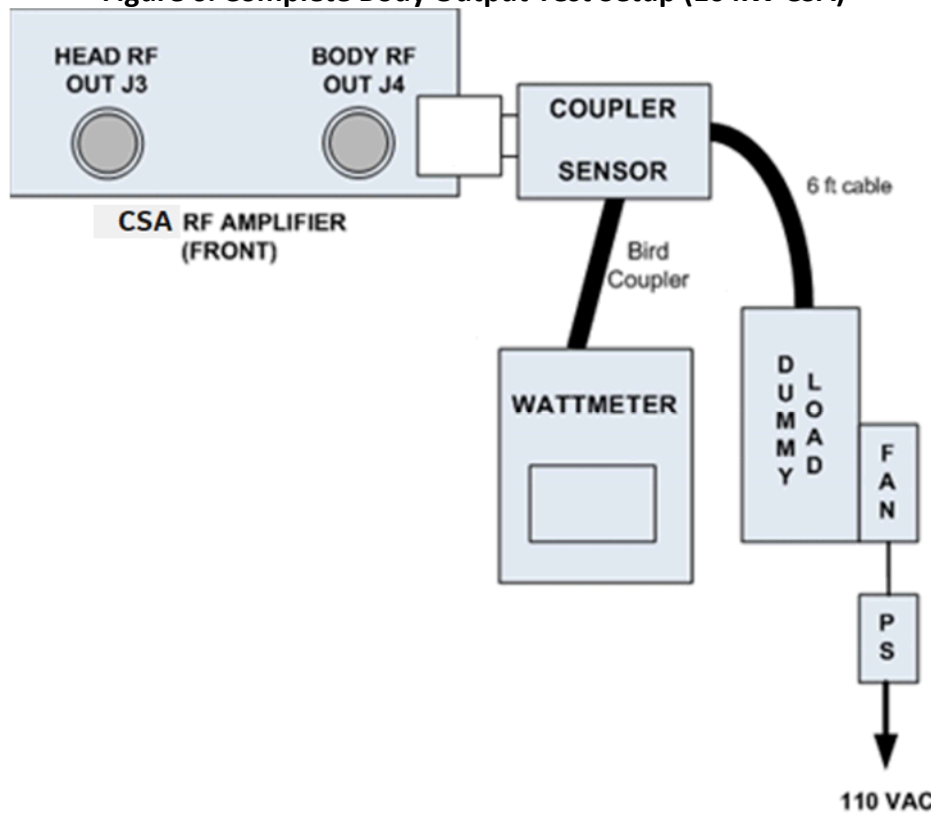
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Figure 6. Complete Body Output Test Setup (10 kW CSA)



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- 5) Plug in the wattmeter to use a service outlet or wall power (batteries are often not charged). Connect one end of the wattmeter cable to the sensor connector. Connect the other end to the coupler.

DANGER

Do not apply RF power to the dummy load without its fan in operation. Doing so will result in catastrophic damage to the dummy load due to overheating.

- 6) Plug the fan power cable for the dummy load into service outlet or wall power. **Never** use this dummy load without its fan running.

Figure 7. Plug the fan power cable to the dummy load



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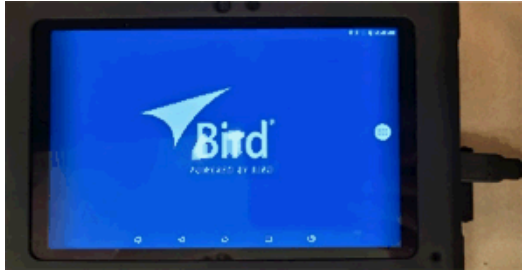
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4. Configuring the wattmeter for body channel

Configure the hardware according to the tool.

Tool	Procedure
Bird 5000-EX 	<i>Configuring the Bird 5000-EX for RF Body Calibration</i> Please refer Step.5
Bird 5000-XT 	<i>Configuring the Bird 5000-XT for RF Body Calibration</i> Please refer Step.6

Tool	Procedure
Bird TT800V 	<i>Configuring the Bird TT800V wattmeter</i> Please refer Step.7

5. Setting up Bird 5000-EX Wattmeter to Measure Forward Body Maximum Power

Figure 8. Bird 5000-EX Wattmeter



- 1) Press the **ON** button on the wattmeter. (If the coupler and sensor are not correctly attached to the amplifier you will see a “No Sensor” indication.)
- 2) Select the arrow **▲** button under Scale.
- 3) Enter **50000w/50kw** to set the range.
- 4) Select the **ENTER** button.

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- 5) Check the setting by selecting the arrow button under Scale.
- 6) Press the **ENTER** button to exit.
- 7) Select the arrow button under **Fwd Units**. Make sure the scale reads **W** (or kW on some wattmeters). If the scale reads dBm, select the arrow button under **Fwd Units** again to toggle the scale to read **W** (Watts) or kW (kiloWatts) on the TOP, forward power scale. Ignore the BOTTOM, reflected power scale value. It is not used during maximum power calibration.
- 8) Select the **SHIFT** button.
- 9) Select the arrow button under Setup.

This displays the sensor type configured in the wattmeter. This procedure uses sensor type 43, which is optimized for peak power measurement.

- 10) Verify that the output reads **43**, and then press the **ENTER** button.

Figure 9. Wattmeter Display Set to Read



- 11) Press the **ENTER** button.
- 12) Press the arrow **▲** button under **Type** and then toggle to change the unit of measurement to **Peak**.
- 13) All kits require an offset to ensure accurate measurement. Locate the Element Offset Factors card in the kit.

Find the offset for “1.5T Value Body Output.”

Figure 10. Example Element Offset Factors Card

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Rf Power Measurement Quick Reference Card for GE3T Kit
Element Offset Factors

Kit Serial Number: 213201609
Calibration Date: 2/23/2023

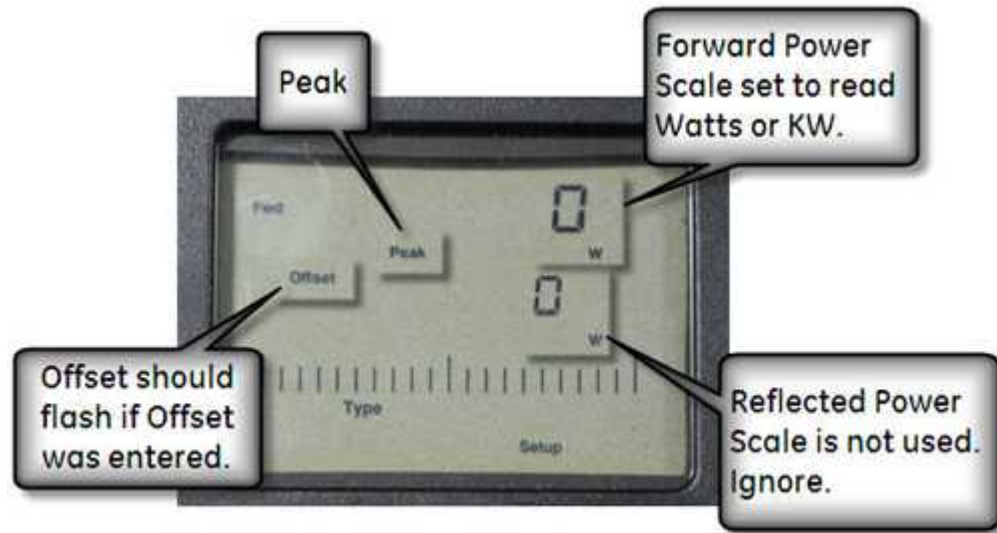
Description	Frequency (MHz)	Power Level (Watts)	Element	Offset
3 T G3 or DVMR Body Output	127.72	35000	4300A374-3	-0.0223
3 T Dual Drive Body Output	127.72	15000	4300A374-3	-0.4982
3 T G3 Head Output	127.72	4000	4300A374-6	-0.0116
3 T DVMR Head Output	127.72	4500	4300A374-6	-0.0769
3 T e2 Body Output	127.72	25000	4300A374-3	-0.0975
3 T e2 Head Output	127.72	3300	4300A374-6	0.1632
1.5 T Body Output	63.86	10000	4300A374-2	-0.0168
1.5 T Value Body Output	63.86	10000	4300A374-2	-0.1833
1.5 T Head Output	63.86	2000	4300A374-5	-0.0335
3 T MNS 4kW Output	51.70	4000	4300A374-5	-0.2174
3 T MNS 8kW Output	51.70	8000	4300A374-2	-0.3384
3 T MNS 2.5kW Output	51.70	2500	4300A374-5	-0.0613

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- 14) Press the **Offset** button on the wattmeter. Enter the numerical value from the offset card and press the **Enter** button.

NOTE: To enter a negative value, first enter the numerical value, then the minus symbol (-).

Figure 11. Wattmeter Set to Display Peak Power



- 15) Proceed to **Step 8. Adjusting RF Amplifier Output Power for Body Mode.**

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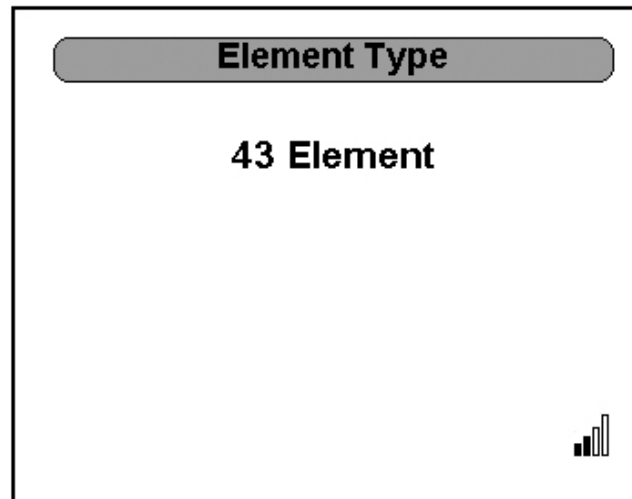
6. Setting up Bird 5000-XT Wattmeter to Measure Forward Body Maximum Power

Figure 12. Bird 5000-XT Wattmeter



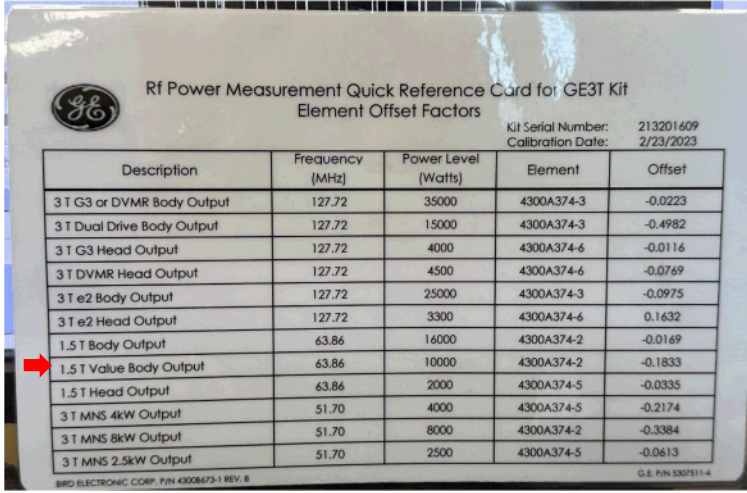
- 1) Press the **OK** button on the wattmeter.
 - 2) Press the **MENU** button. Press the **▼** button to **SCALE MENU**, and then press **OK**.
 - 3) Press **50000w/50kw** to set the range, and then press **OK**.
 - 4) Check the setting by pressing **▼** to **SCALE MENU**, and then press **OK**.
 - 5) Press **OK** to exit. Press **◀** to return to measurement mode.
 - 6) Press **▲** to make sure the scale reads **W** (or **uW** or **kW** on some wattmeters). If the scale reads **dBm**, press **▲** again to toggle the scale to read **W** (or **uW** or **kW** on the TOP, forward power scale. Ignore the BOTTOM, reflected power scale value. It is not used during maximum power calibration.
 - 7) Press **MENU**.
 - 8) Press **▼** to select **ELEMENT TYPE**, and then press **OK**.
- This displays the sensor type configured in the wattmeter. This procedure uses sensor type 43, which is optimized for peak power measurement.
- 9) Verify that the output reads **43**, and then press **OK**.

Figure 13. Wattmeter Set to Read



- 10) Press ◀ to return to measurement mode.
- 11) All kits require an offset to ensure accurate measurement. Locate the Element Offset Factors card in the kit.

Figure 14. Example Element Offset Factors Card



Description	Frequency (MHz)	Power Level (Watts)	Element	Offset
3 T G3 or DVMR Body Output	127.72	35000	4300A374-3	-0.0223
3 T Dual Drive Body Output	127.72	15000	4300A374-3	-0.4982
3 T G3 Head Output	127.72	4000	4300A374-6	-0.0116
3 T DVMR Head Output	127.72	4500	4300A374-6	-0.0769
3 T e2 Body Output	127.72	25000	4300A374-3	-0.0975
3 T e2 Head Output	127.72	3300	4300A374-6	0.1632
1.5 T Body Output	63.86	16000	4300A374-2	-0.0169
1.5 T Value Body Output	63.86	10000	4300A374-2	-0.1833
1.5 T Head Output	63.86	2000	4300A374-5	-0.0335
3 T MNS 4kW Output	51.70	4000	4300A374-5	-0.2174
3 T MNS 8kW Output	51.70	8000	4300A374-2	-0.3384
3 T MNS 2.5kW Output	51.70	2500	4300A374-5	-0.0613

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G.E. P/N 5307511-4

- 12) Press the **MENU** button on the wattmeter. Select **OFFSET**, and then press **OK**. Enter in the numerical value that was displayed on the kit's offset card and press **OK**. (The wattmeter rounds to three significant digits.) Press ◀ to return to measurement mode.

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NOTE: To enter a negative value, first enter the numerical value, and then the minus symbol (–).

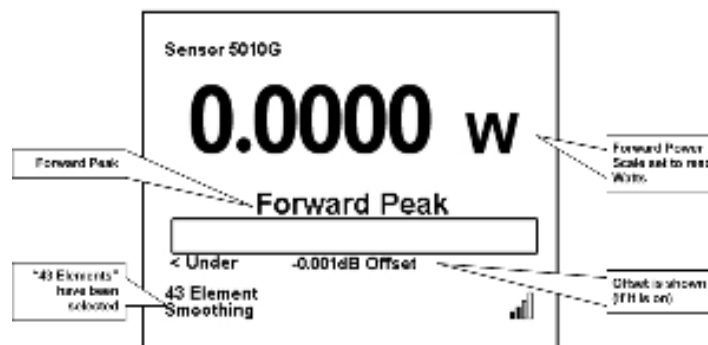
13) Press **MENU**. Press ▼ until **SMOOTHING** is selected, and then press **OK**.

14) The wattmeter should display **4 Samples**. Press **OK**.

15) Press ◀ to return to measurement mode.

16) Press ◀ to **Forward Peak** in measurement mode.

Figure 15. Wattmeter Set to Display Peak Power



17) Proceed to **Step 8. Adjusting RF Amplifier Output Power for Body Mode**.

7. Setting up Bird TT800V Wattmeter to Measure Forward Body Maximum Power

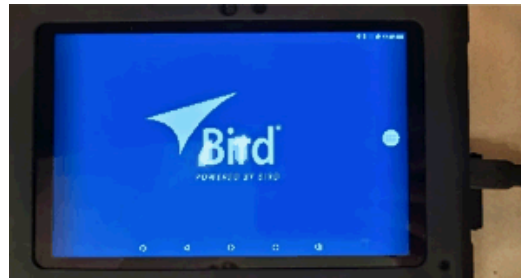
1) Press the power button on the top right section of the Bird TT800V wattmeter unit to turn on the wattmeter.

Figure 16. Bird TT800V wattmeter power button



2) Plug the M5014G sensor into the USB port located on the right side of the wattmeter.

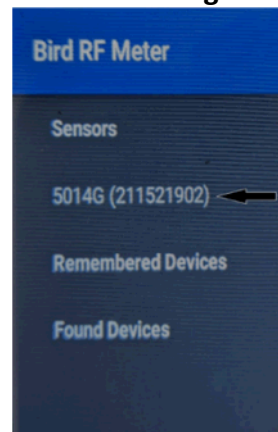
Figure 17. Bird TT800V wattmeter connected to the M5014G sensor



Result: The application should automatically open after plugging in the sensor and show a list of available sensors.

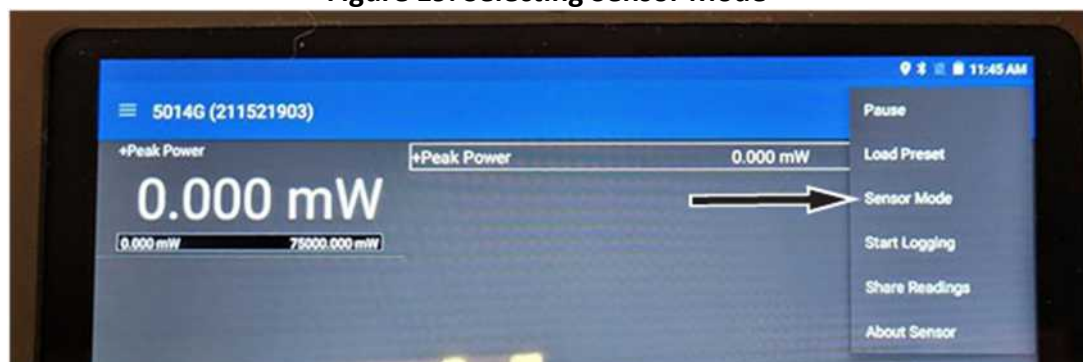
- 3) Select the **M5014G** sensor.

Figure 18. Selecting M5014G



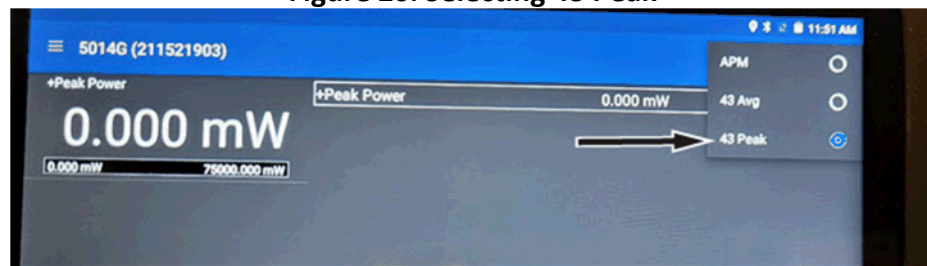
- 4) Click the elipsis on the top right corner of the screen and select **Sensor Mode**.

Figure 19. Selecting Sensor Mode



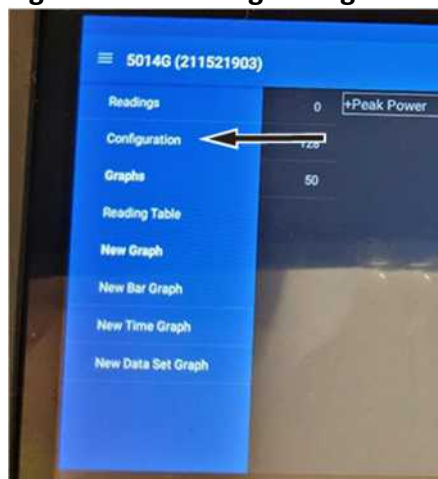
- 5) Select **43 Peak**.

Figure 20. Selecting 43 Peak



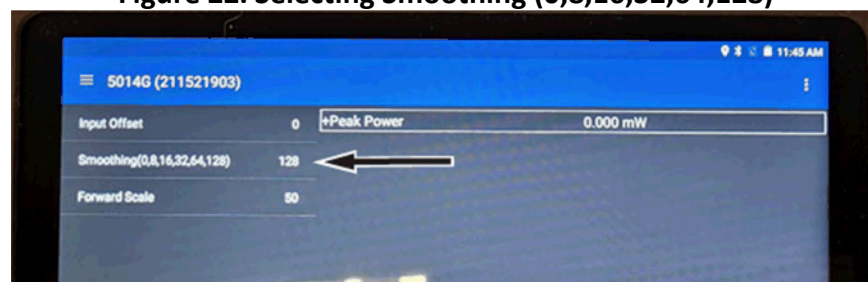
- 6) Click the menu bar on the top left corner of the screen and select **Configuration**.

Figure 21. Selecting Configuration



- 7) Select **Smoothing (0,8,16,32,64,128)**.

Figure 22. Selecting Smoothing (0,8,16,32,64,128)

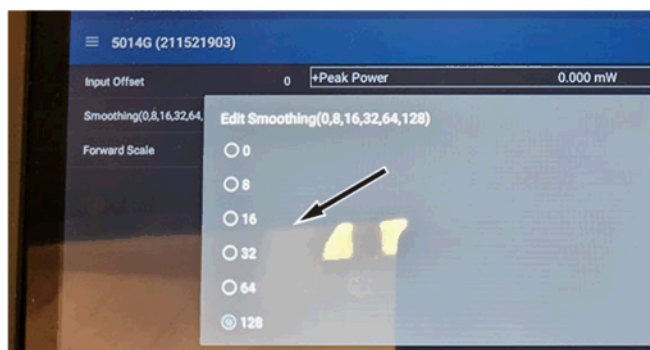


- 8) Select a smoothing level value.

Figure 23. Specifying a smoothing level

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- 9) Locate the Element Offset Factors card in the kit and find the offset for the procedure being run.

This output is used to make sure of an accurate measurement.

Figure 24. Example Element Offset Factors card

Description	Frequency (MHz)	Power Level (Watts)	Element	Offset
3 T G3 or DVMR Body Output	127.72	35000	4300A374-3	-0.0223
3 T Dual Drive Body Output	127.72	15000	4300A374-3	-0.4982
3 T G3 Head Output	127.72	4000	4300A374-6	-0.0116
3 T DVMR Head Output	127.72	4500	4300A374-6	-0.0769
3 T e2 Body Output	127.72	25000	4300A374-3	-0.0975
3 T e2 Head Output	127.72	3300	4300A374-6	0.1632
1.5 T Body Output	63.86	16000	4300A374-2	-0.0169
1.5 T Value Body Output	63.86	10000	4300A374-2	-0.1833
1.5 T Head Output	63.86	8000	4300A374-5	-0.0226
3 T MNS 4kW Output	51.70	4000	4300A374-5	-0.2174
3 T MNS 8kW Output	51.70	8000	4300A374-2	-0.3384
3 T MNS 2.5kW Output	51.70	2500	4300A374-5	-0.0613

- 10) Click the menu bar on the top left corner of the screen and select **Configuration**.

- 11) Select **Input Offset** and enter the value found on the Element Offset Factors card.

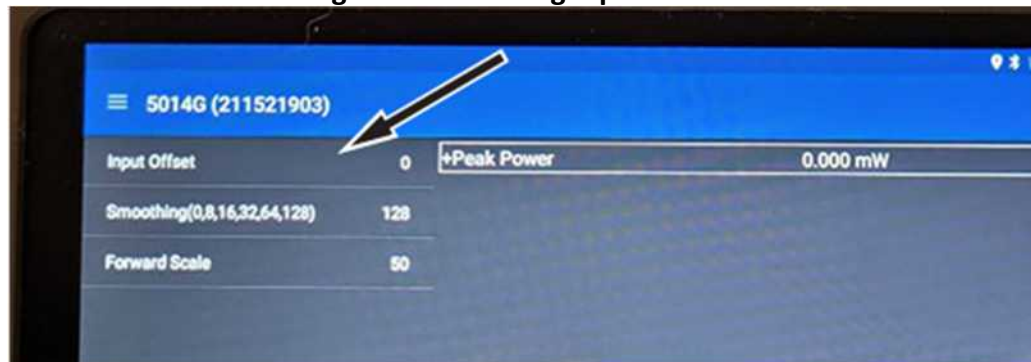
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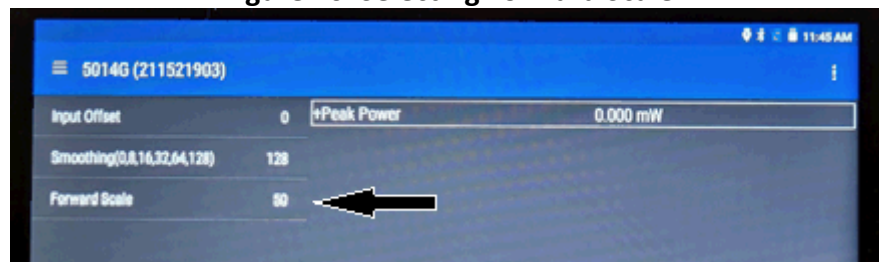
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Figure 25. Selecting Input Offset



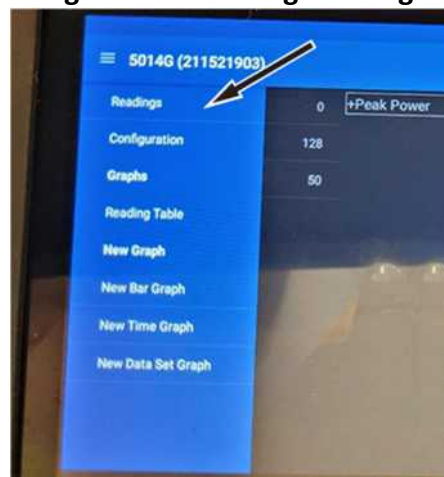
12) Select **Forward Scale** and enter **50000w/5kw** for body channel to set the range.

Figure 26. Selecting Forward Scale



13) Click the menu bar on the top left corner of the screen and select **Readings**.

Figure 27. Selecting Readings

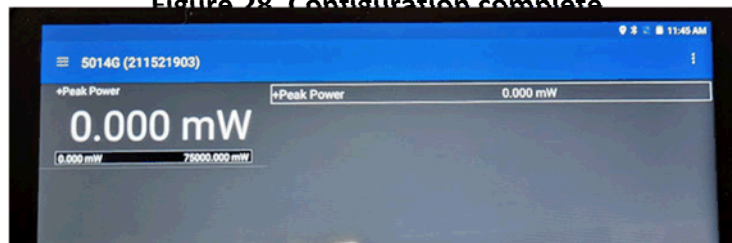


Results:

The meter is now ready to calibrate the RF amplifier output power.

NOTE: The units will automatically update when power is being measured.

Figure 28. Configuration complete



8. Adjusting RF amplifier output power for body channel

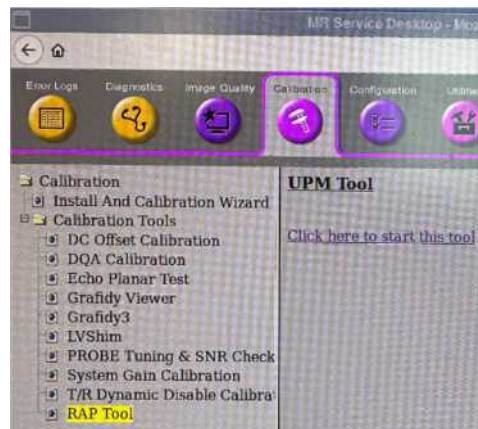
- 1) Place the RF Enable switch on the exciter (located on the left side of the System Cabinet) to the enable (up) position.

Figure 29. RF ENB to enable (up) position



- 2) Open service browser. Go to calibration RAP/UPM Tool Choose "Click here to start this tool".

Figure 30. RF ENB to enable (up) position

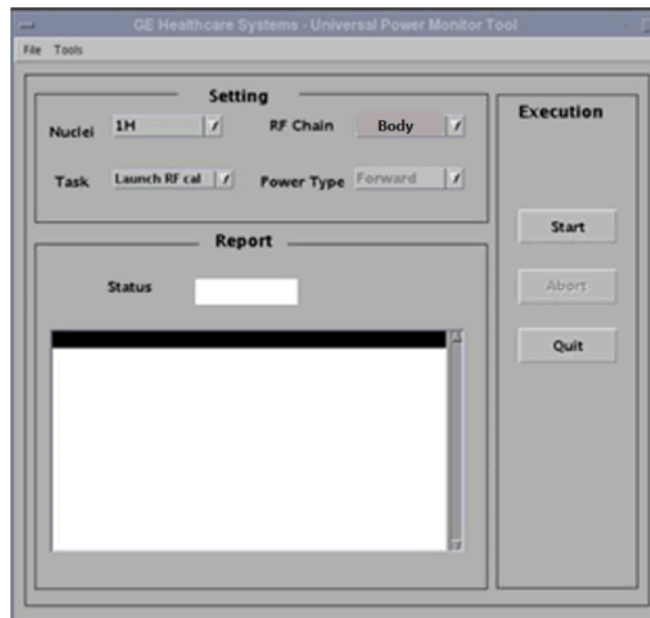


- 3) The calibration tool will open. Please choose task as "Launch RF cal" and RF Chain as "Body".

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(For System running before SV29.1/SV25.4)



(For System running SV29.1/SV25.4 or later)



4) Click **Start**.

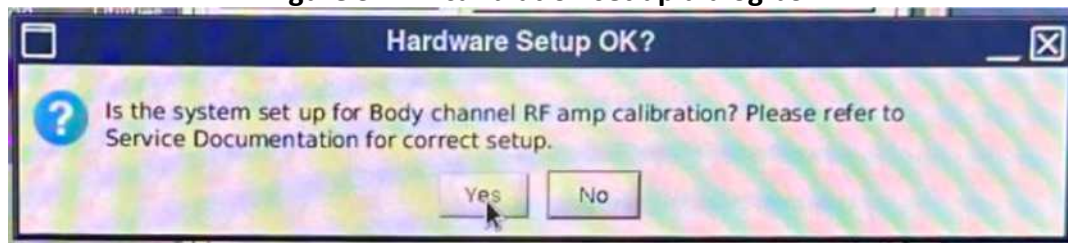
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- 5) When a dialog box shows asking if the system is properly set up, click **Yes**.

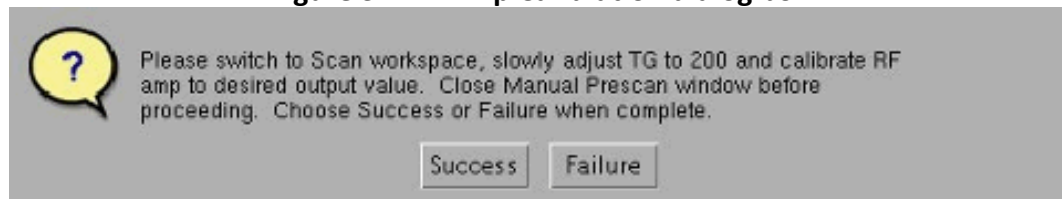
Figure 31. RF calibration set up dialog box



Result: The system sets the proper sequence for RF calibration. This process takes about 1 minute.

- 6) When a dialog box shows, do not click **Success** or **Failure** at this time. Instead, navigate to the *Scan Desktop*.

Figure 32. RF Amp Calibration dialog box



- 7) From the *Scan Desktop*, click on the patient icon to navigate to the *Scan Workplace*.

Figure 33. Patient Icon



- 8) Then slowly move the **Transmit Gain (TG)** slide to **100**.

NOTE: For all the **Transmit Gain (TG)** adjustment steps given in this procedure, after the TG slider is set and released in the *Prescan Looping* window, make sure that the setting is retained by the system (for example the slider does not move back to some other TG setting) prior to executing the next step in the procedure. If the setting was not retained, move the TG slider to the desired setting again and make sure of the setting.

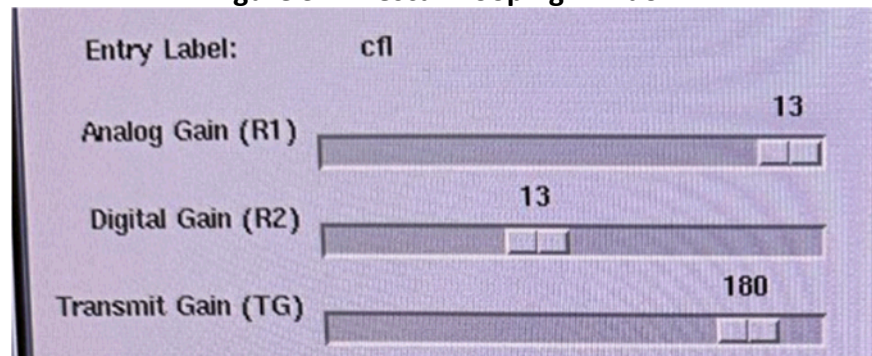
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- 9) In the *Prescan Looping* window, slowly increase the **TG** to **180**.

Figure 34. Prescan Looping Window

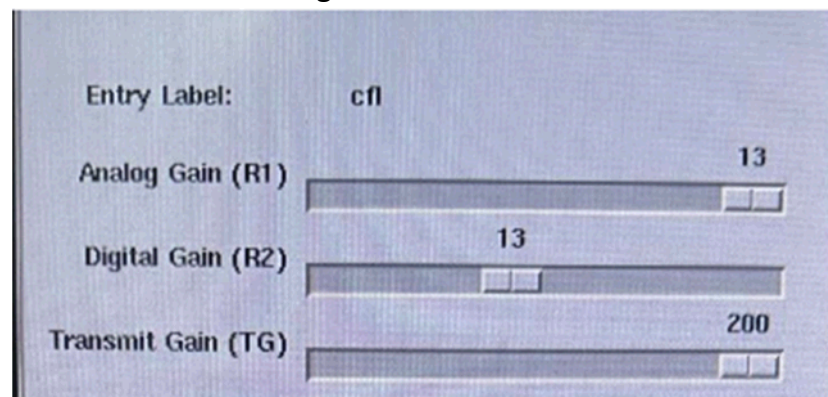


- 10) Make sure everything is functioning normally and there is a reading on the wattmeter.

NOTE: If the manual prescan must be stopped and restarted, you must relaunch the calibration sequence using the UPM/RAP tool. Click **Failure** in the calibration dialog box and then click **Start** on the UPM/RAP tool with the same menu selections to relaunch the calibration sequence.

- 11) Slowly increase the **TG** to **200**.

Figure 35. TG to 200



NOTE: Allow 10 seconds for the meter reading to stabilize before making the next adjustment.

NOTE: (For SV29.1/SV25.4 or later) Enter the Config File Manager and select *prescan_config*. Find and record the value given for the max value of TG. If the value is not 200, change it to 200, save changes, and then reboot. Refer to *Troubleshooting RF Amplifier Peak Power Trips* for details. You must do this before you do UPM calibration.

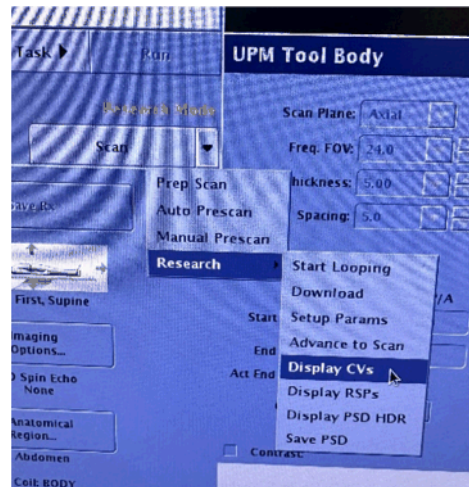
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- 12) If the wattmeter reading is unstable, click **Done**, then select [Research Options], then choose Display CVs.

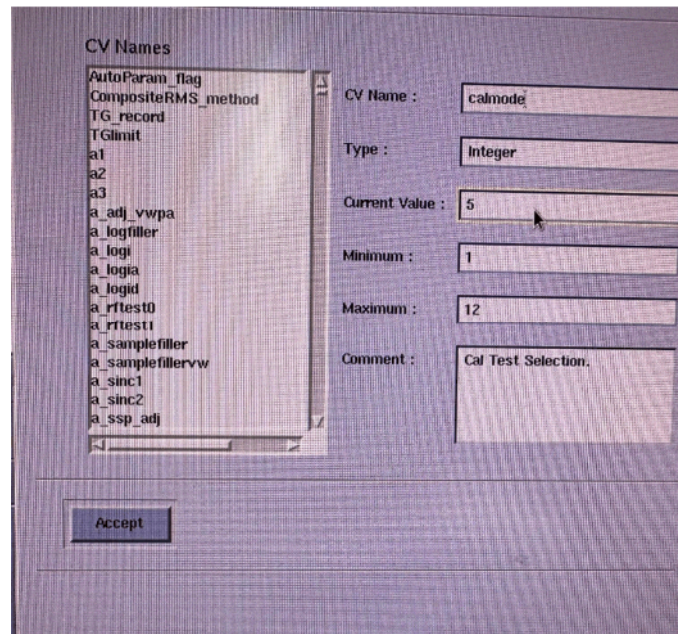
Figure 36. Choose Display CVs



- 13) In the CV Name window, type “calmode”, and then hit enter. Set current value to **5**, then click **Accept**.

NOTE: This step will ensure a stable reading on the wattmeter.

Figure 37. Set current value to 5



- 14) Select [Research Options], then choose **Download**.

Figure 38. Choose Download



- 15) Select [Manual PreScan], then select **SCAN TR (R1/R2)**.

Figure 39. Select Manual PreScan

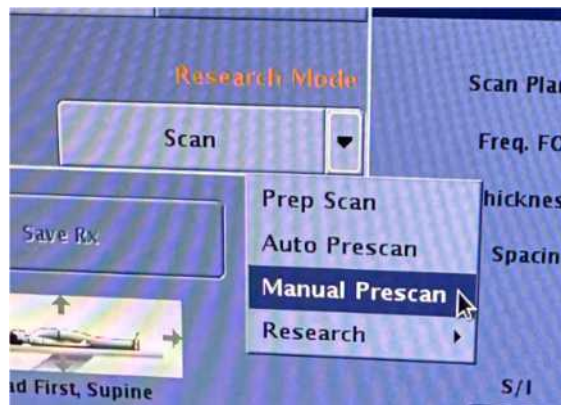
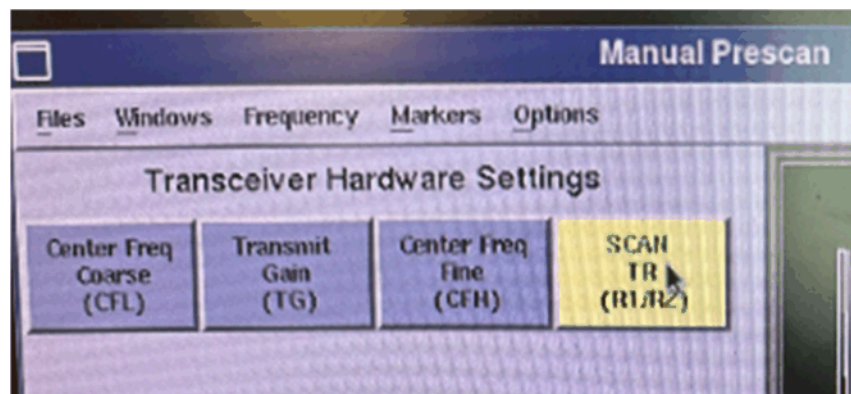


Figure 40. Select TR SCAN TR (R1/R2)



16) Check the wattmeter to see if the reading is 10 kW $\pm$ 500 watts.

Figure 41. Example in BIRD wattmeter



If yes, continue to the next step.

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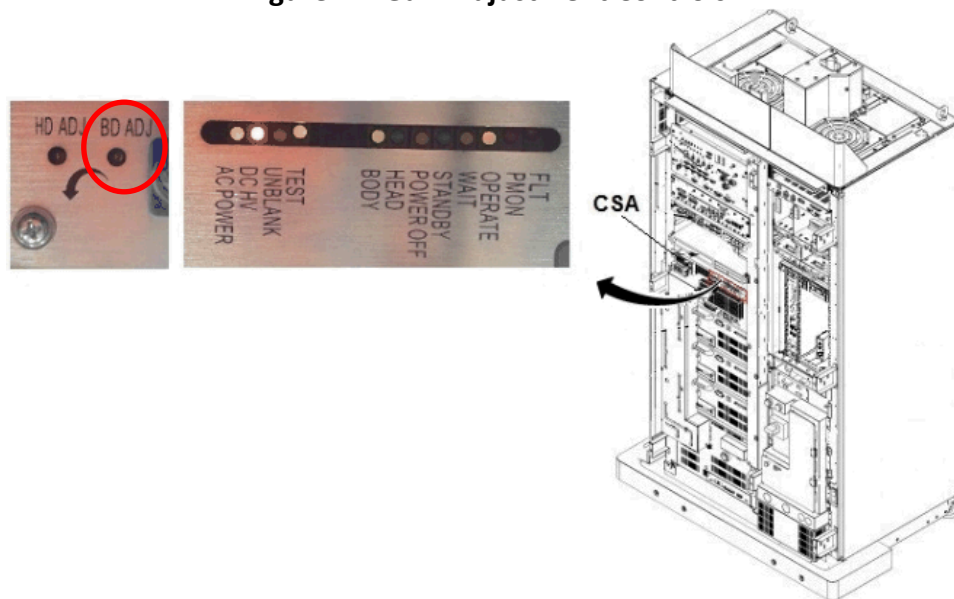
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If no, slowly adjust the gain on the front panel of the RF amp until the wattmeter reads 10 kW $\pm$ 500 watts.

Use Small Flat Head Screwdriver Rotate the single **BDY adjustment encoder** on the RF amplifier front panel and, if necessary, increase the TG in small increments until the wattmeter reads 10 kW $\pm$ 500 watts when the TG is 200. Only adjust while the amplifier is actively pulsing. If output does not change, cycle amplifier power, wait 10 minutes, and then repeat the adjustment again.

Figure 42. Gain Adjustment Controls



17) After the RF output is calibrated to specification, reduce the **TG** to 0 (zero) and click **Done**.

18) From the *Scan Desktop*, click on the **CSA** icon.

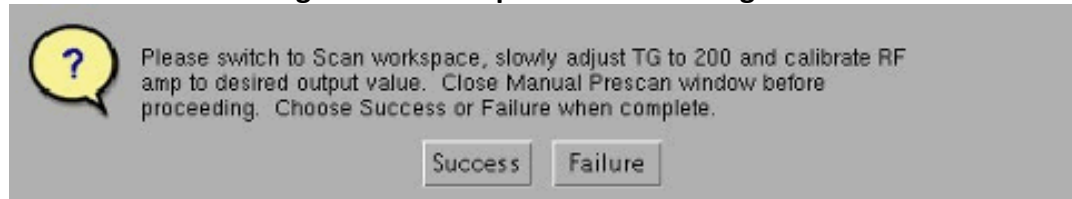
Figure 43. CSA icon



19) If the RF output is within specification, click **Success** in the **RF Amp Calibration** dialog box.

NOTE: The calibration dialog box may be hidden behind the *ICW* window. Move the *ICW* window to the left or right to see the *RF Amp Calibration* dialog box.

Figure 44. RF Amp Calibration dialog box



20) Click **Yes** to make sure that the RF output calibration was successful.

Result: After a few seconds, PASS shows in the status area of the RF Amp & Power Monitor tool.

Figure 45. Confirm Output Power

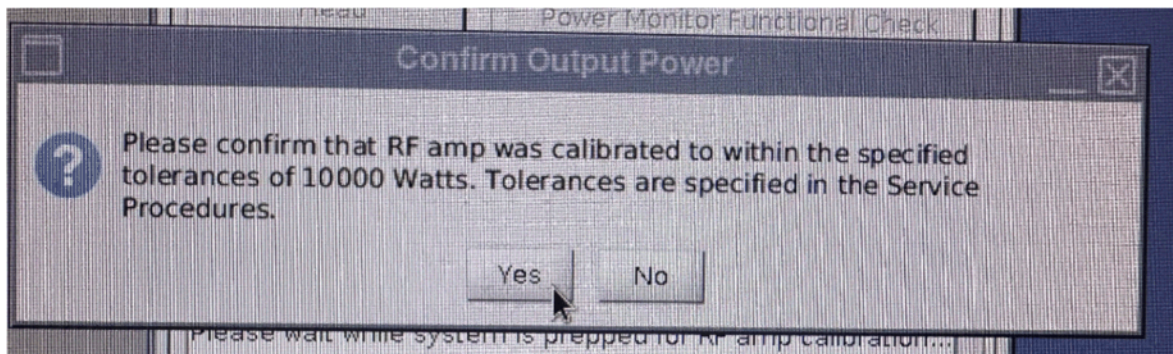
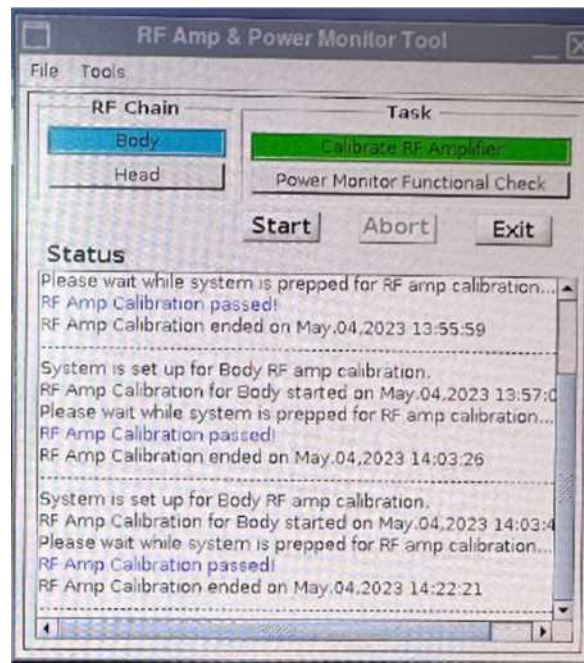


Figure 46. Sample of PASS status of RF Amp & Power Monitor tool.



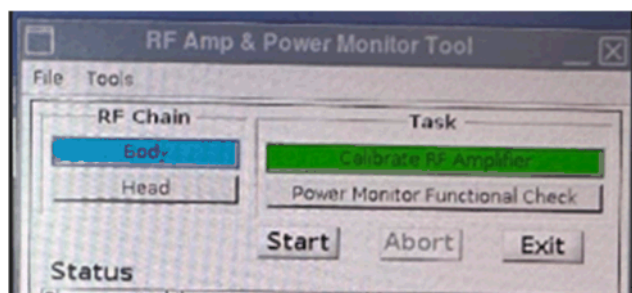
9. RAP/UPM Functional Check for Body Mode

After completing the UPM calibration, continue with the UPM functional check.

From the UPM Tool, make the following selections:

- 1) Select Power Monitor Functional Check, then click **Start**.

Figure 47. Select Power Monitor Functional Check

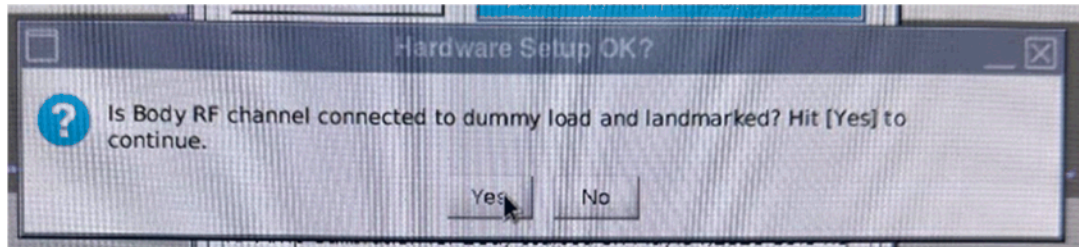


- 2) Select **Yes** when the prompts ask if the hardware is set up correctly.

Figure 48. Hardware Setup prompts ask

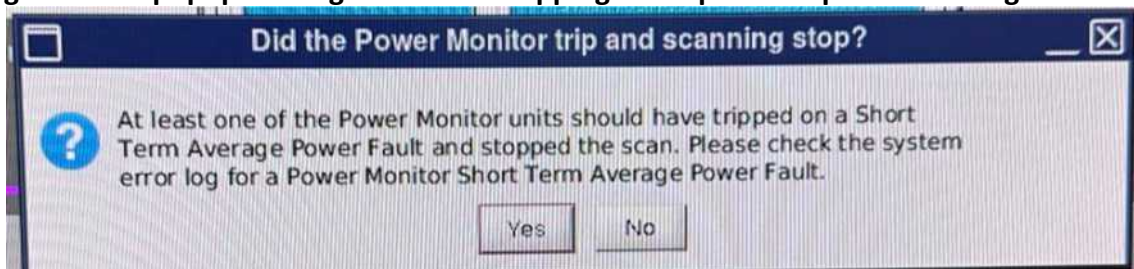
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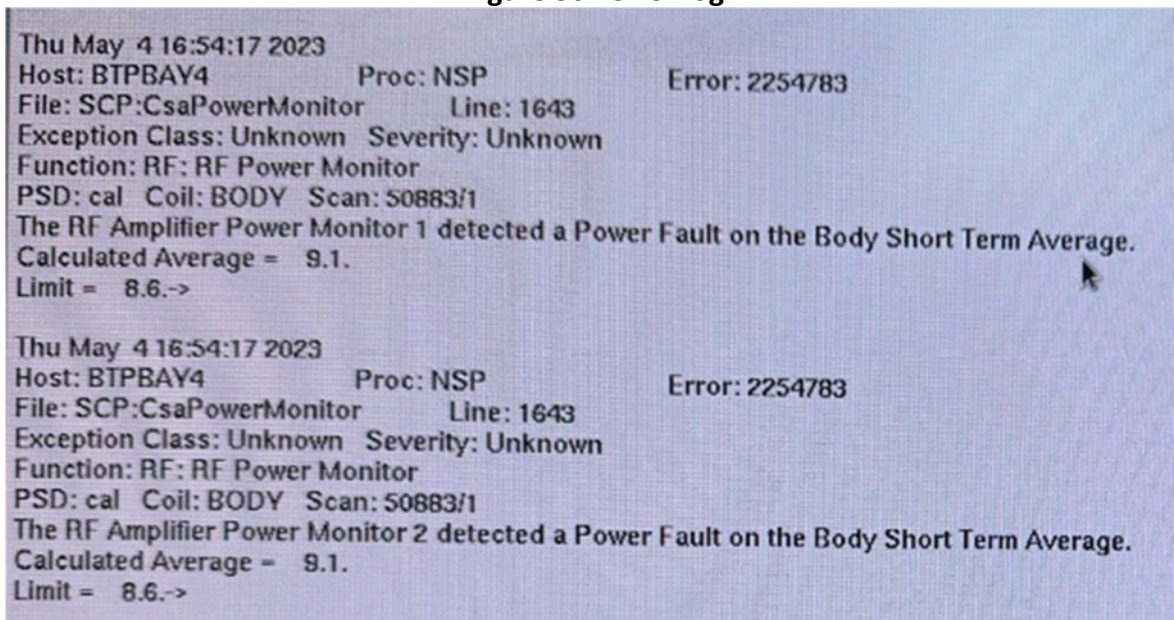
- 3) Prompts are displayed during test execution, including a popup message about the UPM tripping and a power trip error message. The tool performs three tests before the host trip. This takes 2 to 3 minutes before any user interaction is required.

Figure 49. A popup message about the tripping and a power trip error message



- 4) From the CSD, open the error log. Verify the error log contains the message that the UPM has **“detected a power fault on Body Short Term Average.”** Select Yes if an error message is displayed.

Figure 50. error log



- 5) After completing the calibration and check of the body channel, continue to set and calibrate the RF amplifier for head mode **Disabling T/R, DD, and RF Input for Head Mode.**

10. Disabling T/R, DD, and RF Input for Head Mode

DANGER

FATAL ELECTRIC SHOCK HAZARD.

the RF Amplifier is not disabled.

to prevent fatal electric shock, ensure the RF Amplifier is disabled while configuring the calibration tools.

- 1) Place the RF Enable switch on the DTX Exciter, located in the top of the System cabinet, into the **Disable (down)** position.

Figure 51. RF ENB Switch



- 2) Disconnect the coupler from **J4** on the front panel of the RF amplifier and connect the body output cable to **J4**.

11. Setting up RF Coupler for Head Calibration

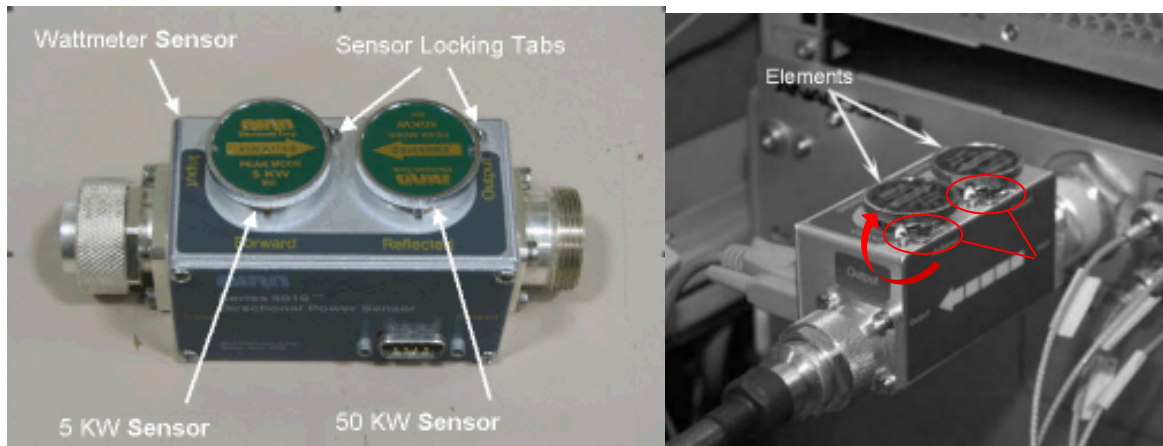
Insert the **5kW** element into the **Forward** sensor port (green 4300A374-5). Lock the element in place using the locking tab on the sensor. The **Reflected** port is not used during this procedure.

Placing the **50kW** element (green 4300A374-2) into the Reflected port is just a safety measure and is not used during body calibration.

NOTE: Make sure the arrows on the elements point in opposite directions.

NOTE: Make sure that the two elements are plugged into the port and locked with buckles, otherwise it may cause inaccurate readings or no value display on the power meter.

Figure 52. RF Coupler Setup for Head Mode



12. Wattmeter and Dummy Load Setup for Head Channel (CSA)

DANGER

FATAL ELECTRIC SHOCK HAZARD!
THE RF AMPLIFIER IS NOT DISABLED.
TO PREVENT FATAL ELECTRIC SHOCK, ENSURE THE RF AMPLIFIER IS DISABLED WHILE CONFIGURING THE CALIBRATION TOOLS.

WARNING

ELECTRICAL/RF BURN HAZARD!
RF BURNS Can Occur AS A RESULT OF DISCONNECTING HELIAX CABLES WHILE THE SYSTEM IS SCANNING.
PREVENT POSSIBLE RF BURNS WHEN DISCONNECTING HELIAX CABLES FROM THE RF AMPLIFIER BY VERIFYING THAT THE SYSTEM IS NOT MANUALLY PRESCANNING OR SCANNING. VERIFY THAT THE SCAN DESKTOP ICON DISPLAYS THE "IDLE" MESSAGE AND THE RF ENABLE SWITCH IS IN THE "OFF" POSITION.

NOTICE

Possible equipment damage. Do not connect/disconnect the wattmeter sensor while the meter is ON. Always turn the meter OFF before plugging/unplugging the sensor.

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- 7) Verify that the scanner is idle. Confirm that the **RF ENB switch** on the DTX1 exciter is in the **Disable (down)** position.
- 8) Disconnect the head output cable from **J3** on the front panel of the RF amplifier.
- 9) Connect the input of the coupler with **an adapter** to **J3** on the front panel of the RF amplifier.

Figure 53. Connect coupler with an adapter



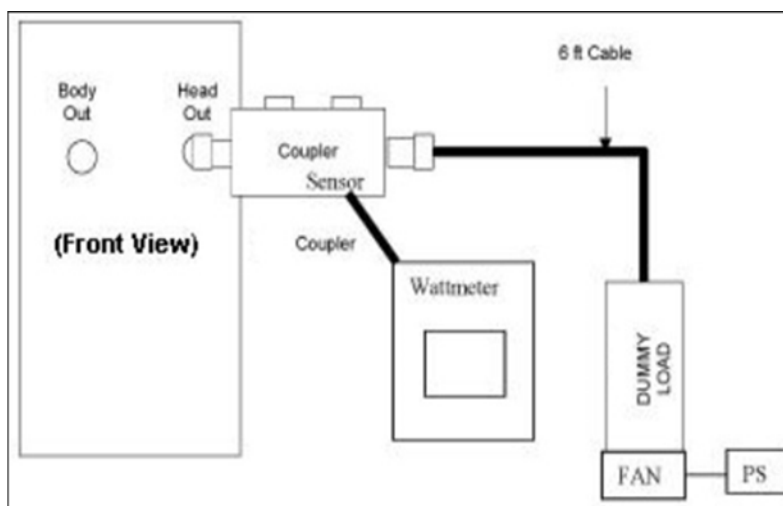
WARNING

ELECTRIC SHOCK RISK!

Failure to connect the dummy load correctly could result in electric shock.
Pay close attention to the connections to the dummy load.

- 10) Connect the output of the RF coupler to the dummy load.

Figure 54. Complete Body Output Test Setup (10 kW CSA)



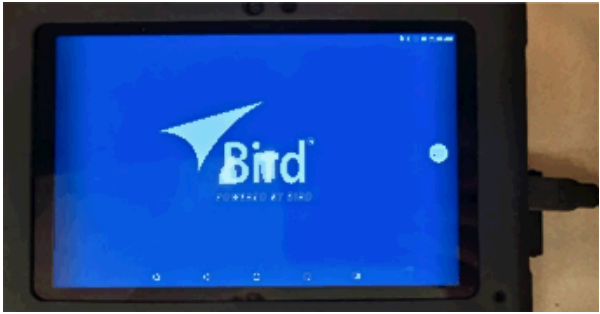
13. Configuring the wattmeter for Head channel

Configure the hardware according to the tool.

Tool	Procedure
Bird 5000-EX 	<i>Configuring the Bird 5000-EX for RF Body Calibration</i> Please refer Step.14

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Tool	Procedure
Bird 5000-XT 	<i>Configuring the Bird 5000-XT for RF Body Calibration</i> Please refer Step.15
Bird TT800V 	<i>Configuring the Bird TT800V wattmeter</i> Please refer Step.16

14. Setting up Bird 5000-EX Wattmeter to Measure Forward Head Maximum Power

Figure 55. Bird 5000-EX Wattmeter

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- 1) Press the **ON** button on the wattmeter. (If the coupler and sensor are not correctly attached to the amplifier you will see a “No Sensor” indication.)
- 2) Select the arrow **▲** button under Scale.
- 3) Enter **5000w/5kw** to set the range.
- 4) Select the **ENTER** button.
- 5) Check the setting by selecting the arrow button under Scale.
- 6) Press the **ENTER** button to exit.
- 7) Select the arrow button under **Fwd Units**. Make sure the scale reads **W** (or **kW** on some wattmeters). If the scale reads dBm, select the arrow button under **Fwd Units** again to toggle the scale to read **W** (Watts) or **kW** (kiloWatts) on the TOP, forward power scale. Ignore the BOTTOM, reflected power scale value. It is not used during maximum power calibration.
- 8) Select the **SHIFT** button.
- 9) Select the arrow button under Setup.

This displays the sensor type configured in the wattmeter. This procedure uses sensor type 43, which is optimized for peak power measurement.

- 10) Verify that the output reads **43**, and then press the **ENTER** button.

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Figure 56. Wattmeter Display Set to Read



- 11) Press the **ENTER** button.
- 12) Press the arrow **▲** button under **Type** and then toggle to change the unit of measurement to **Peak**.
- 13) All kits require an offset to ensure accurate measurement. Locate the Element Offset Factors card in the kit.

Find the offset for “1.5T Value Body Output.”

Figure 57. Example Element Offset Factors Card

 **Rf Power Measurement Quick Reference Card for GE3T Kit**
Element Offset Factors

Kit Serial Number: 213201609
Calibration Date: 2/23/2023

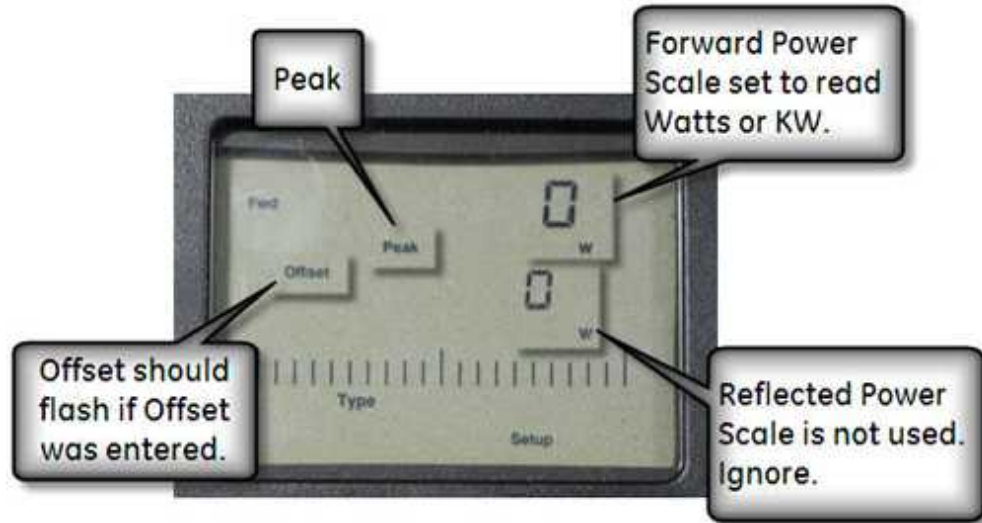
Description	Frequency (MHz)	Power Level (Watts)	Element	Offset
3 T G3 or DVMR Body Output	127.72	35000	4300A374-3	-0.0223
3 T Dual Drive Body Output	127.72	15000	4300A374-3	-0.4982
3 T G3 Head Output	127.72	4000	4300A374-6	-0.0116
3 T DVMR Head Output	127.72	4500	4300A374-6	-0.0769
3 T e2 Body Output	127.72	25000	4300A374-3	-0.0975
3 T e2 Head Output	127.72	3300	4300A374-6	0.1632
1.5 T Body Output	63.86	16000	4300A374-2	-0.0169
1.5 T Value Body Output	63.86	10000	4300A374-2	-0.1833
1.5 T Head Output	63.86	2000	4300A374-5	-0.0335
3 T MNS 4kW Output	51.70	4000	4300A374-3	-0.2174
3 T MNS 8kW Output	51.70	8000	4300A374-2	-0.3384
3 T MNS 2.5kW Output	51.70	2500	4300A374-5	-0.0613

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- 14) Press the **Offset** button on the wattmeter. Enter the numerical value from the offset card and press the **Enter** button.

NOTE: To enter a negative value, first enter the numerical value, then the minus symbol (-).

Figure 58. Wattmeter Set to Display Peak Power



15) Proceed to **Step 17. Adjusting RF Amplifier Output Power for Head Mode.**

15. Setting up Bird 5000-XT Wattmeter to Measure Forward Head Maximum Power

Figure 59. Bird 5000-XT Wattmeter



1) Press the **OK** button on the wattmeter.

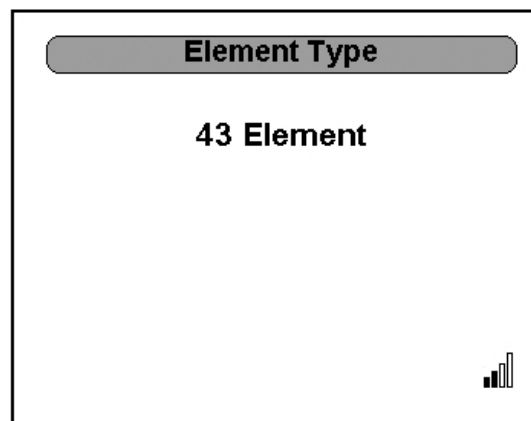
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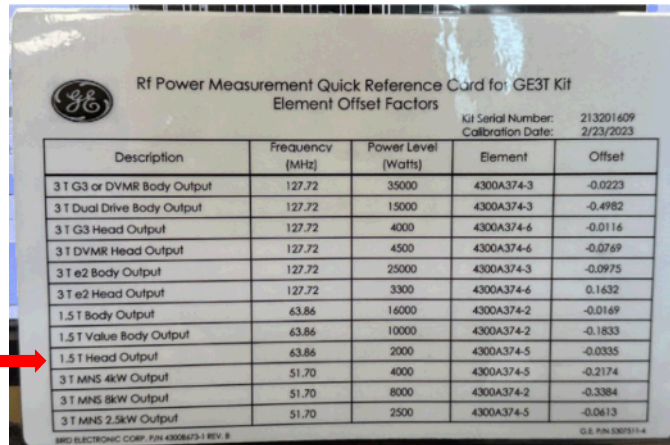
- 2) Press the **MENU** button. Press the ▼ button to **SCALE MENU**, and then press **OK**.
 - 3) Press **5000w/5kw** to set the range, and then press **OK**.
 - 4) Check the setting by pressing ▼ to **SCALE MENU**, and then press **OK**.
 - 5) Press **OK** to exit. Press ◀ to return to measurement mode.
 - 6) Press ▲ to make sure the scale reads **W** (or **uW** or **kW** on some wattmeters). If the scale reads **dBm**, press ▲ again to toggle the scale to read **W** (or **uW** or **kW** on the TOP, forward power scale. Ignore the BOTTOM, reflected power scale value. It is not used during maximum power calibration.
 - 7) Press **MENU**.
 - 8) Press ▼ to select **ELEMENT TYPE**, and then press **OK**.
- This displays the sensor type configured in the wattmeter. This procedure uses sensor type 43, which is optimized for peak power measurement.
- 9) Verify that the output reads **43**, and then press **OK**.

Figure 60. Wattmeter Set to Read



- 10) Press ◀ to return to measurement mode.
- 11) All kits require an offset to ensure accurate measurement. Locate the Element Offset Factors card in the kit.

Figure 61. Example Element Offset Factors Card



Description	Frequency (MHz)	Power Level (Watts)	Element	Offset
3 T G3 or DVMR Body Output	127.72	35000	4300A374-3	-0.0223
3 T Dual Drive Body Output	127.72	15000	4300A374-3	-0.4982
3 T G3 Head Output	127.72	4000	4300A374-6	-0.0116
3 T DVMR Head Output	127.72	4500	4300A374-6	-0.0769
3 T e2 Body Output	127.72	25000	4300A374-3	-0.0975
3 T e2 Head Output	127.72	3300	4300A374-6	0.1632
1.5 T Body Output	63.86	16000	4300A374-2	-0.0169
1.5 T Valve Body Output	63.86	10000	4300A374-2	-0.1833
1.5 T Head Output	63.86	2000	4300A374-5	-0.0335
3 T MNS 4kW Output	51.70	4000	4300A374-5	-0.2174
3 T MNS 8kW Output	51.70	8000	4300A374-2	-0.3384
3 T MNS 2.5kW Output	51.70	2500	4300A374-5	-0.0613

12) Press the **MENU** button on the wattmeter. Select **OFFSET**, and then press **OK**. Enter in the numerical value that was displayed on the kit's offset card and press **OK**. (The wattmeter rounds to three significant digits.) Press **◀** to return to measurement mode.

NOTE: To enter a negative value, first enter the numerical value, and then the minus symbol (–).

13) Press **MENU**. Press **▼** until **SMOOTHING** is selected, and then press **OK**.

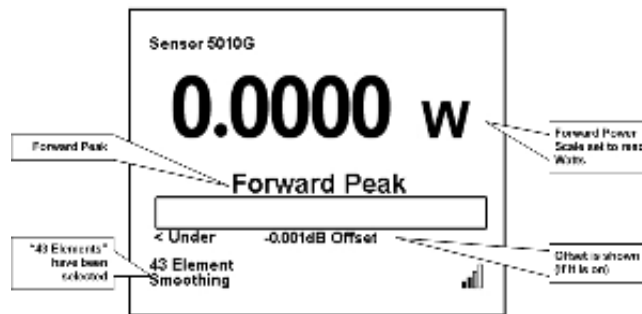
14) To verify that **SMOOTHING** has been set, press **▼** until **SMOOTHING** is selected, and then press **OK**.

15) The wattmeter should display **4 Samples**. Press **OK**.

16) Press **◀** to return to measurement mode.

17) Press **◀** to **Forward Peak** in measurement mode.

Figure 62. Wattmeter Set to Display Peak Power



18) Proceed to **Step 17. Adjusting RF Amplifier Output Power for Head Mode.**

16. Setting up Bird TT800V Wattmeter to Measure Forward Head Maximum Power

- 1) Press the power button on the top right section of the Bird TT800V wattmeter unit to turn on the wattmeter.

Figure 63. Bird TT800V wattmeter power button



- 2) Plug the M5014G sensor into the USB port located on the right side of the wattmeter.

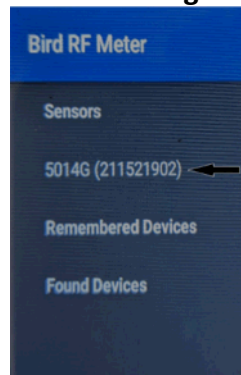
Figure 64. Bird TT800V wattmeter connected to the M5014G sensor



Result: The application should automatically open after plugging in the sensor and show a list of available sensors.

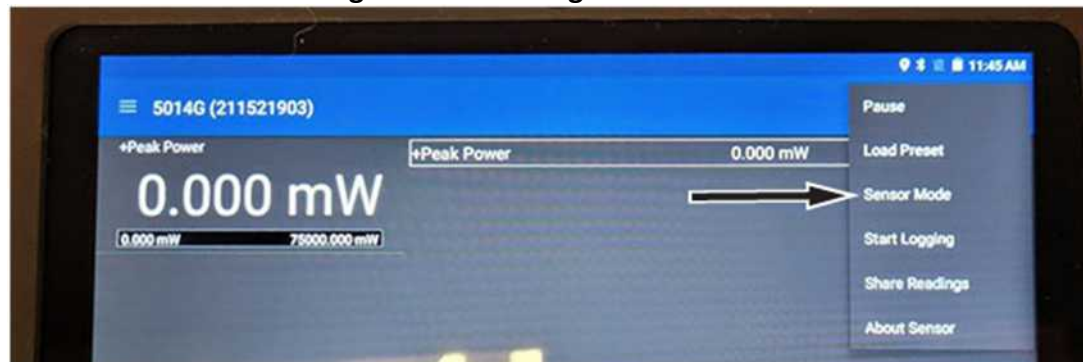
- 3) Select the **M5014G** sensor.

Figure 65. Selecting M5014G



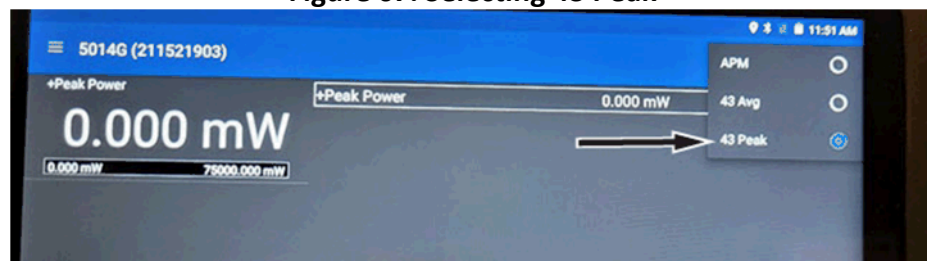
- 4) Click the elipsis on the top right corner of the screen and select **Sensor Mode**.

Figure 66. Selecting Sensor Mode



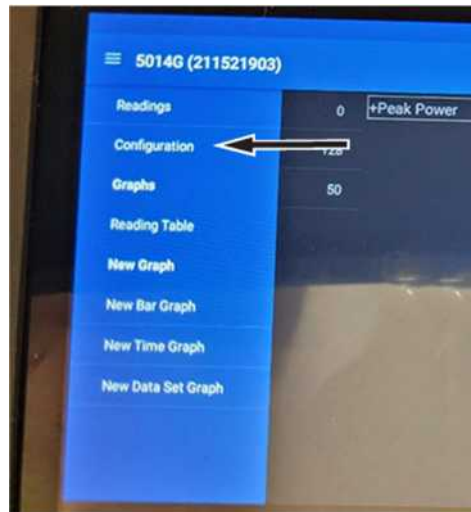
- 5) Select **43 Peak**.

Figure 67. Selecting 43 Peak



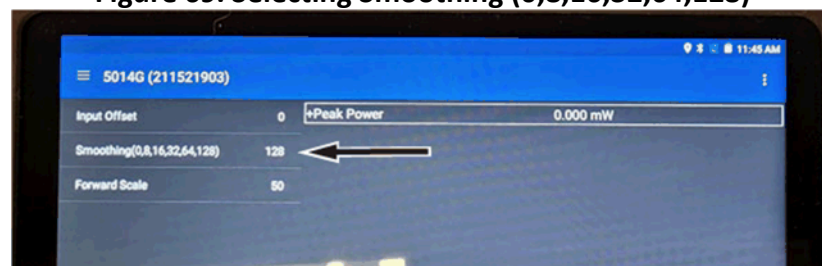
- 6) Click the menu bar on the top left corner of the screen and select **Configuration**.

Figure 68. Selecting Configuration



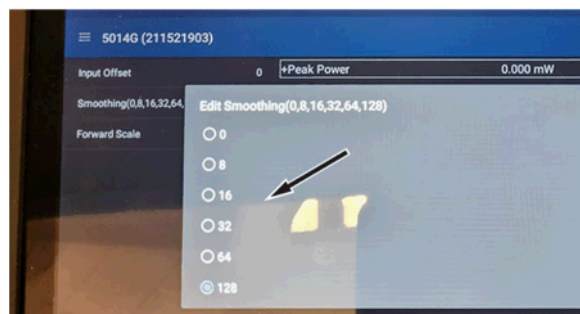
- 7) Select **Smoothing (0,8,16,32,64,128)**.

Figure 69. Selecting Smoothing (0,8,16,32,64,128)



- 8) Select a smoothing level value.

Figure 70. Specifying a smoothing level



- 9) Locate the Element Offset Factors card in the kit and find the offset for the procedure being run.

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This output is used to make sure of an accurate measurement.

Figure 71. Example Element Offset Factors card

Rf Power Measurement Quick Reference Card for GE3T Kit
Element Offset Factors

Kit Serial Number: 213201609
Calibration Date: 2/23/2023

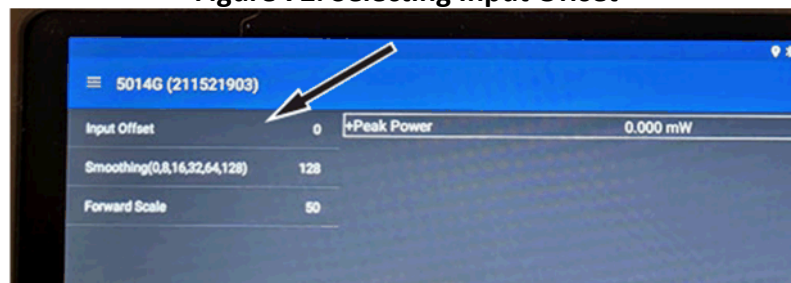
Description	Frequency (MHz)	Power Level (Watts)	Element	Offset
3 T G3 or DVMR Body Output	127.72	35000	4300A374-3	-0.0223
3 T Dual Drive Body Output	127.72	15000	4300A374-3	-0.4982
3 T G3 Head Output	127.72	4000	4300A374-6	-0.0116
3 T DVMR Head Output	127.72	4500	4300A374-6	-0.0769
3 T e2 Body Output	127.72	25000	4300A374-3	-0.0975
3 T e2 Head Output	127.72	3300	4300A374-6	0.1632
1.5 T Body Output	63.86	16000	4300A374-2	-0.0169
1.5 T Value Body Output	63.86	10000	4300A374-2	-0.1833
1.5 T Head Output	63.86	2000	4300A374-5	-0.0335
3 T MNS 4kW Output	51.70	4000	4300A374-5	-0.2174
3 T MNS 8kW Output	51.70	8000	4300A374-2	-0.3384
3 T MNS 2.5kW Output	51.70	2500	4300A374-5	-0.0613

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G.E. P/N 5207511-4

10) Click the menu bar on the top left corner of the screen and select **Configuration**.

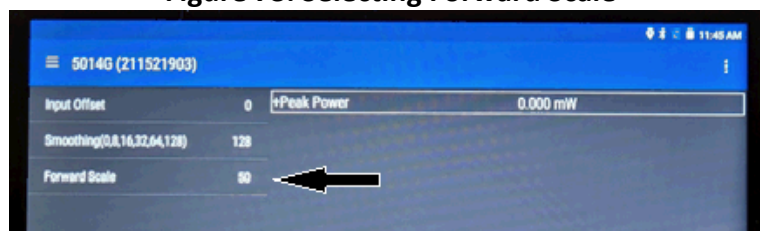
11) Select **Input Offset** and enter the value found on the Element Offset Factors card.

Figure 72. Selecting Input Offset



12) Select **Forward Scale** and enter **5000w/5kw** for head channel to set the range.

Figure 73. Selecting Forward Scale



13) Click the menu bar on the top left corner of the screen and select **Readings**.

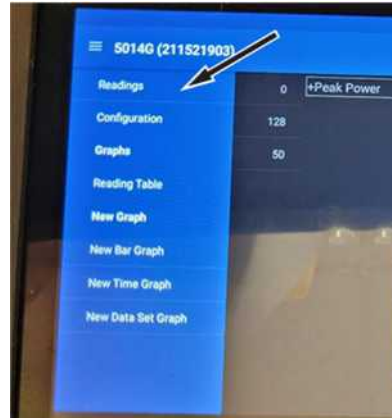
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Figure 74. Selecting Readings

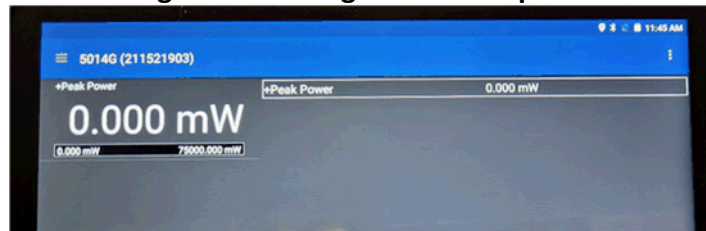


Results:

The meter is now ready to calibrate the RF amplifier output power.

NOTE: The units will automatically update when power is being measured.

Figure 75. Configuration complete



17. Adjusting RF amplifier output power for Head channel

- 1) Place the RF Enable switch on the exciter (located on the left side of the System Cabinet) to the **enable (up)** position.

Figure 76. RF ENB in enable (up) position

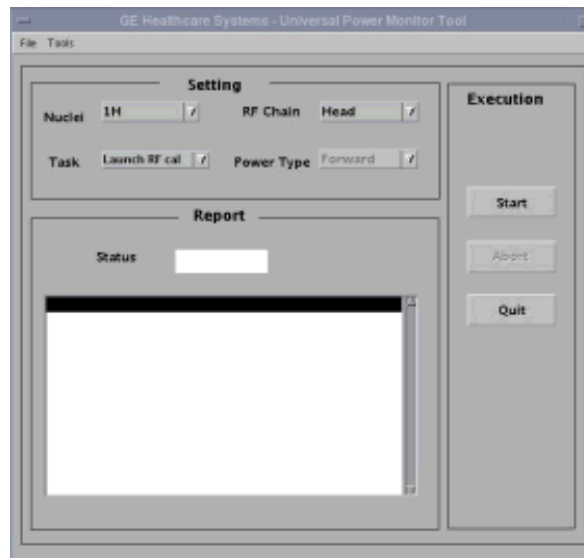


- 2) Open service browser. Go to calibration UPM Tool Choose "Click here to start this tool"
- 3) The calibration tool will open. Please choose task as "Launch RF cal" and RF Chain as "Head". Click "Start".

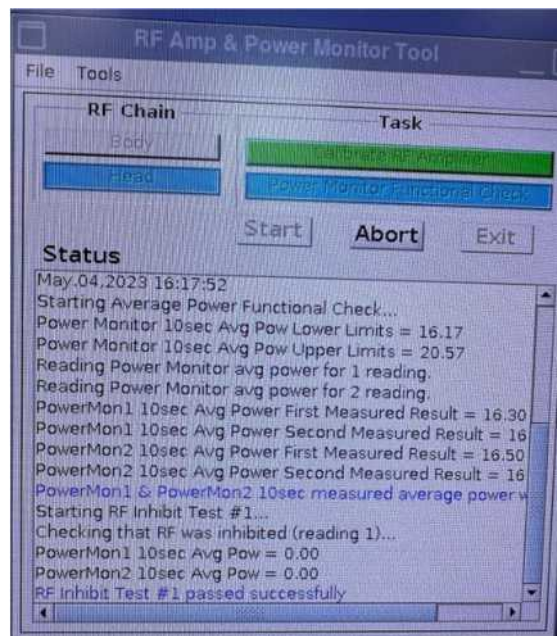
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(For System running before SV29.1/SV25.4)



(For System running SV29.1/SV25.4 or later)



- 4) Click **Start**.
- 5) When a dialog box shows asking if the system is properly set up, click **Yes**.

Figure 77. RF calibration set up dialog box

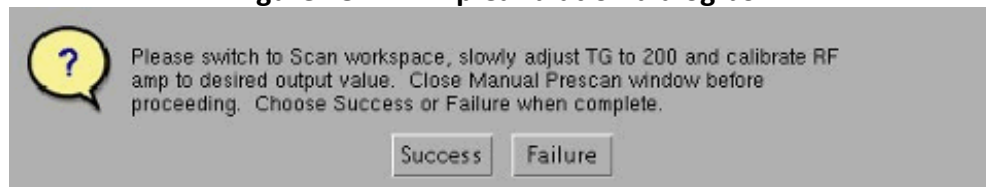


Result:

The system sets the proper sequence for RF calibration. This process takes about 1 minute.

- 6) When a dialog box shows, do not click **Success** or **Failure** at this time. Instead, navigate to the *Scan Desktop*.

Figure 78. RF Amp Calibration dialog box



- 7) From the *Scan Desktop*, click on the patient icon to navigate to the *Scan Workplace*.

Figure 79. Patient Icon



- 8) Then slowly move the **Transmit Gain (TG)** slide to **100**.

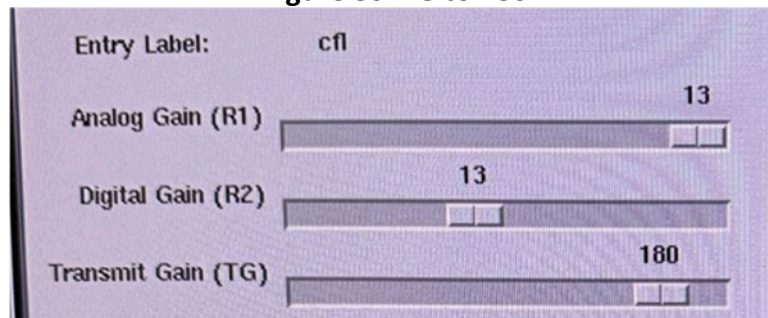
NOTE: For all the **Transmit Gain (TG)** adjustment steps given in this procedure, after the TG slider is set and released in the *Prescan Looping* window, make sure that the setting is retained by the system (for example the slider does not move back to some other TG setting) prior to executing the next step in the procedure. If the setting was not retained, move the TG slider to the desired setting again and make sure of the setting.

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- 9) In the *Prescan Looping* window, slowly increase the **TG** to **180**.

Figure 80. TG to 180

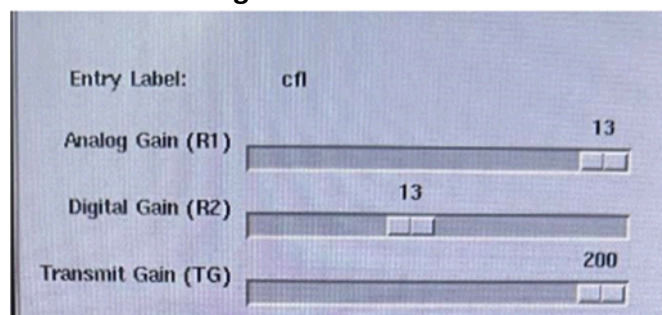


NOTE: Make sure everything is functioning normally and there is a reading on the wattmeter.

NOTE: If the manual prescan must be stopped and restarted, you must relaunch the calibration sequence using the UPM/RAP tool. Click **Failure** in the calibration dialog box and then click **Start** on the UPM/RAP tool with the same menu selections to relaunch the calibration sequence.

- 10) Slowly increase the **TG** to **200**.

Figure 81. TG to 200



NOTE: Allow 10 seconds for the meter reading to stabilize before making the next adjustment.

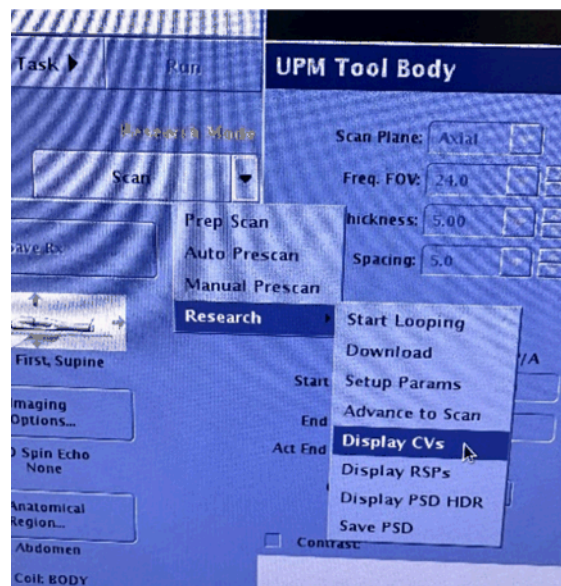
NOTE: (For SV29.1/SV25.4 or later) Enter the Config File Manager and select *prescan_config*. Find and record the value given for the max value of TG. If the value is not 200, change it to 200, save changes, and then reboot. Refer to *Troubleshooting RF Amplifier Peak Power Trips* for details. You must do this before you do UPM calibration.

- 11) If the wattmeter reading is unstable, click **Done**, then select [Research Options], then choose Display CVs.

Figure 82. Choose Display CVs

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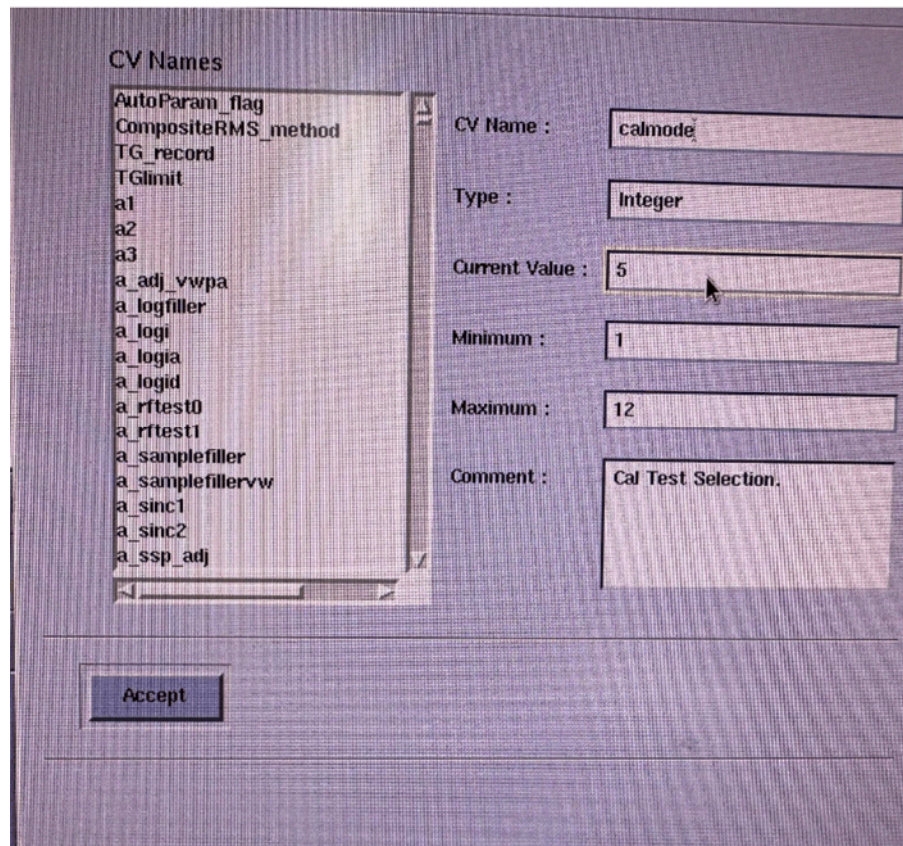
- 12) In the CV Name window, type “calmode”, and then hit enter. Set current value to **5**, then click **Accept**.

NOTE: This step will ensure a stable reading on the wattmeter.

Figure 83. Set current value to 5

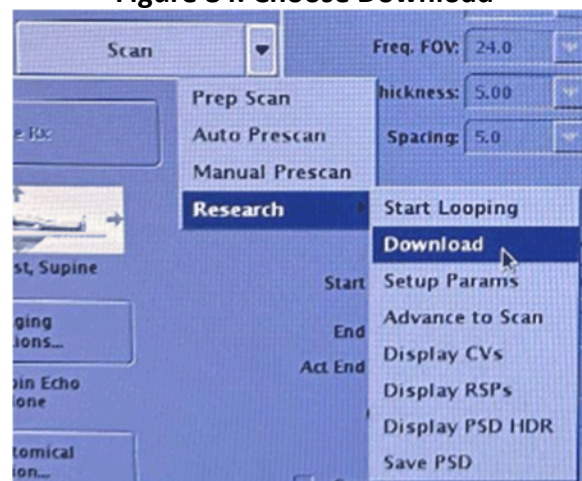
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- 13) Select [Research Options], then choose **Download**.

Figure 84. Choose Download



- 14) Select [Manual PreScan], then select **SCAN TR (R1/R2)**.

Figure 85. Select Manual PreScan

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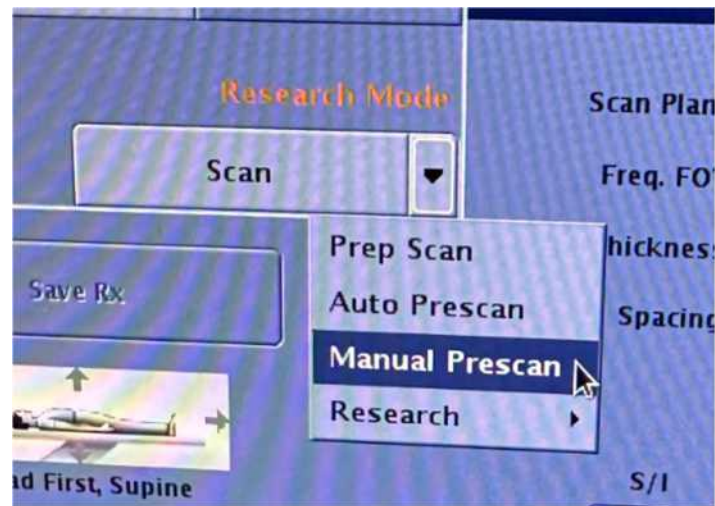
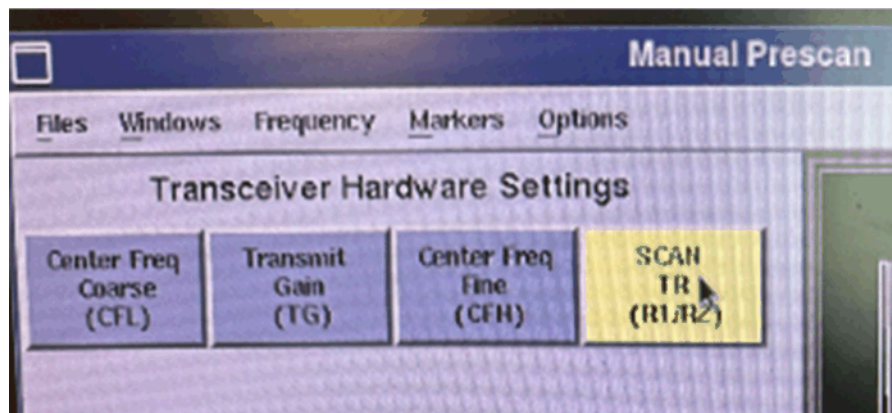


Figure 86. Select TR SCAN TR (R1/R2)



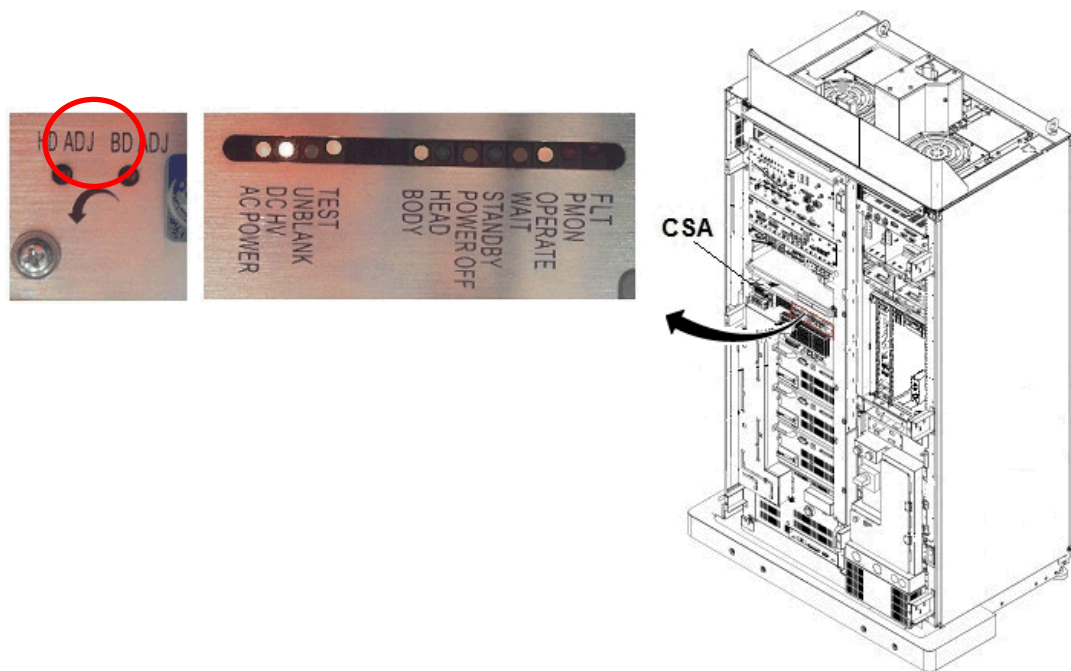
- 15) Check the wattmeter to see if the reading is $2 \text{ kW} \pm 100 \text{ watts}$.
 If yes, continue to the next step.
 If no, slowly adjust the gain on the front panel of the RF amp until the wattmeter reads $2 \text{ kW} \pm 100 \text{ watts}$.

Use Small Flat Head Screwdriver rotate the single **HD adjustment encoder** on the RF amplifier front panel and, if necessary, increase the TG in small increments until the wattmeter reads $2 \text{ kW} \pm 100 \text{ watts}$ when the TG is 200. Only adjust while the amplifier is actively pulsing. If output does not change, cycle amplifier power, wait 10 minutes, and then repeat the adjustment again.

Figure 87. Gain Adjustment Controls

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16) After the RF output is calibrated to specification, reduce the **TG** to 0 (zero) and click **Done**.

17) From the *Scan Desktop*, click on the **CSD** icon.

Figure 88. CSD icon



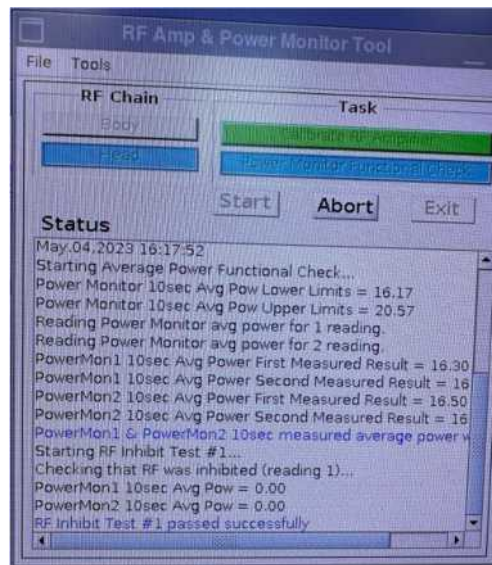
18) If the RF output is within specification, click **Success** in the **RF Amp Calibration** dialog box.

NOTE: The calibration dialog box may be hidden behind the *ICW* window. Move the *ICW* window to the left or right to see the *RF Amp Calibration* dialog box.

19) Click **Yes** to make sure that the RF output calibration was successful.

20) **Result:** After a few seconds, **PASS** shows in the status area of the RF Amp & Power Monitor tool.

Figure 89. Sample of PASS status of RF Amp & Power Monitor tool.



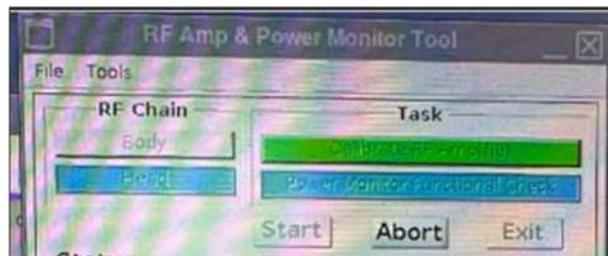
18. RAP/UPM Functional Check for Head Mode

After completing the UPM calibration, continue with the UPM functional check.

From the UPM Tool, make the following selections:

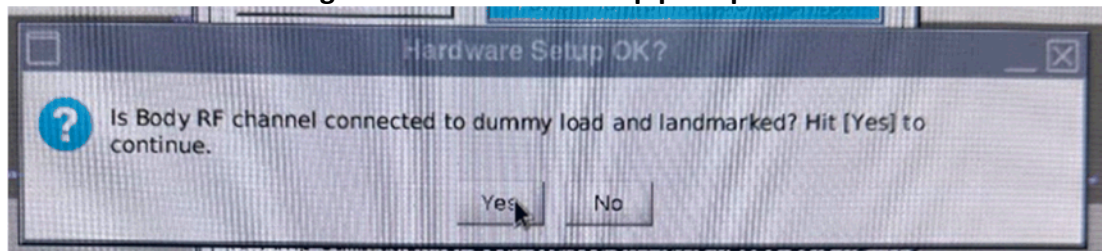
- 1) Select Power Monitor Functional Check, then click **Start**.

Figure 90. Select Power Monitor Functional Check then start



- 2) Select **Yes** when the prompts ask if the hardware is set up correctly.

Figure 91. Hardware Setup prompts ask



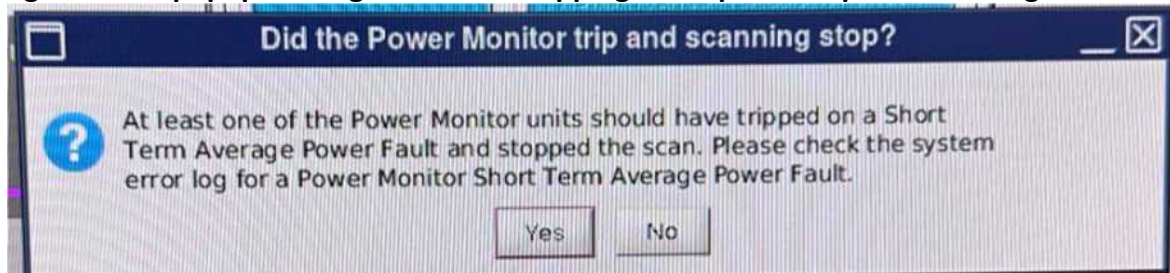
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- 3) Prompts are displayed during test execution, including a popup message about the UPM tripping and a power trip error message. The tool performs three tests before the host trip. This takes 2 to 3 minutes before any user interaction is required.

Figure 92. A popup message about the tripping and a power trip error message



- 4) From the CSD, open the error log. Verify the error log contains the message that the UPM has **“detected a power fault on Head Short Term Average.”** Select Yes if an error message is displayed.

Figure 93. Error log

```

Thu May 4 16:23:31 2023
Host: BTPBAY4          Proc: NSP          Error: 2254783
File: SCP:CsaPowerMonitor Line: 1643
Exception Class: Unknown Severity: Unknown
Function: RF: RF Power Monitor
PSD: cal Coil: HCI Scan: 50883/1
The RF Amplifier Power Monitor 1 detected a Power Fault on the Head Short Term Average.
Calculated Average = 8.7.
Limit = 8.6.->

Thu May 4 16:23:31 2023
Host: BTPBAY4          Proc: NSP          Error: 2254783
File: SCP:CsaPowerMonitor Line: 1643
Exception Class: Unknown Severity: Unknown
Function: RF: RF Power Monitor
PSD: cal Coil: HCI Scan: 50883/1
The RF Amplifier Power Monitor 2 detected a Power Fault on the Head Short Term Average.
Calculated Average = 8.8.
Limit = 8.6.->

```

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19.Finalization

WARNING

ELECTRICAL/RF BURN HAZARD!

RF BURNS Can Occur AS A RESULT OF DISCONNECTING HELIAX CABLES WHILE THE SYSTEM IS SCANNING.

PREVENT POSSIBLE RF BURNS WHEN DISCONNECTING HELIAX CABLES FROM THE RF AMPLIFIER BY VERIFYING THAT THE SYSTEM IS NOT MANUALLY PRESCANNING OR SCANNING. VERIFY THAT THE SCAN DESKTOP ICON DISPLAYS THE "IDLE" MESSAGE AND THE RF ENABLE SWITCH IS IN THE "OFF" POSITION.

NOTICE

Possible equipment damage. Do not connect/disconnect the wattmeter sensor while the wattmeter is ON. Always turn the wattmeter OFF before plugging/unplugging sensor.

- 1) Set the **RF ENB** switch to DISABLE (down).
- 2) Disconnect the RF coupler from **J3** and reconnect the head transmit line.
- 3) Re-enable the TR and DD fault switches on the front of the driver module.
- 4) Set **RF ENB** switch to ENABLE (up).
- 5) Confirm that the other RF amplifier system cables are reconnected to their correct locations for clinical scanning.
- 6) Perform **Doing a TPS reset**.
- 7) Perform **Check Scan**.
- 8) If all the calibrations have been successfully made, perform a **SaveInfo** to save the UPM calibration results Save and Restore Tool.

Revision	Version	Date	Author	Description
1	1	25th-May-2023	Qin Yu (SSO: 503343456)	Initial release

End of Document